CHAPTER 12

Separations and Purifications

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: b94770f451153af12a6f8cbc8f6fbf1cb44c24d3dafd102797f67e42d01ba88b

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch12:exercise:acid-extraction | acid-extraction:stage-1, acid-extraction:stage-2, acid-extraction:stage-3

ch12:exercise:choose-distillation | choose-distillation:stage-1, choose-distillation:stage-2, choose-distillation:stage-3

ch12:exercise:rf-calculation | rf-calculation:stage-1, rf-calculation:stage-2, rf-calculation:stage-3

ch12:exercise:chromatography-match | chromatography-match:stage-1, chromatography-match:stage-2, chromatography-match:stage-3

ch12:lesson:solubility | solubility:rule, solubility:cue, solubility:trap, solubility:app, solubility:step-1, solubility:step-2, solubility:step-3

ch12:lesson:distillation | distillation:rule, distillation:cue, distillation:trap, distillation:app, distillation:step-1, distillation:step-2

ch12:lesson:tlc | tlc:rule, tlc:cue, tlc:trap, tlc:app, tlc:step-1, tlc:step-2

ch12:lesson:specialized | specialized:rule, specialized:cue, specialized:trap, specialized:app, specialized:step-1, specialized:step-2

ch12:molecule:benzoicacid | benzoicacid

ch12:molecule:benzoate | benzoate

ch12:molecule:toluene | toluene

ch12:molecule:glycylglycine | glycylglycine

ch12:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 12.1 Solubility-Based Methods

scope-2: 12.2 Distillation

scope-3: 12.3 Chromatography

Objective: Plan acid–base extraction and crystallization

Objective: Choose distillation conditions

Objective: Calculate Rf and match chromatography methods

Supplied Kaplan curriculum: Chapter 12; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter12b.xhtml

Size exclusion principles: Waters educational primer | https://www.waters.com/nextgen/us/en/education/primers/beginners-guide-to-size-exclusion-chromatography/gpc-basic-chemistry.html

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

solubility | Extraction, filtration, and recrystallization

Change solubility to move a compound

solubility:rule | Core Rules

Liquid–liquid extraction partitions solutes between immiscible phases. Acid/base chemistry can turn a neutral acid or base into a more water-soluble ion. Recrystallization exploits high hot-solvent and low cold-solvent solubility of the desired compound. Hot gravity filtration removes insoluble material; vacuum filtration collects crystals.

Schematic summary: Change solubility to move a compound

Phases — Choose immiscible solvents

Charge — Acid/base chemistry can favor water solubility

Recovery — Track the target layer

solubility:cue | Memory Cues

For an acid: deprotonate to move toward water, then acidify to recover the neutral form.

Schematic summary: Choose the case that applies

Move an acid toward water — Deprotonate to carboxylate

Recover the neutral acid — Acidify its aqueous fraction

solubility:trap | Common Mistakes / MCAT Traps

The organic layer is not always on top; density determines position. Keep track of which layer contains the target. Several smaller extractions can outperform one using the same total solvent volume. A wash removes impurities while retaining the target in its phase.

Schematic summary: Shortcut versus a better check

Avoid — Organic layer is always on top

Check — Density and the target location

solubility:app | MCAT Applications

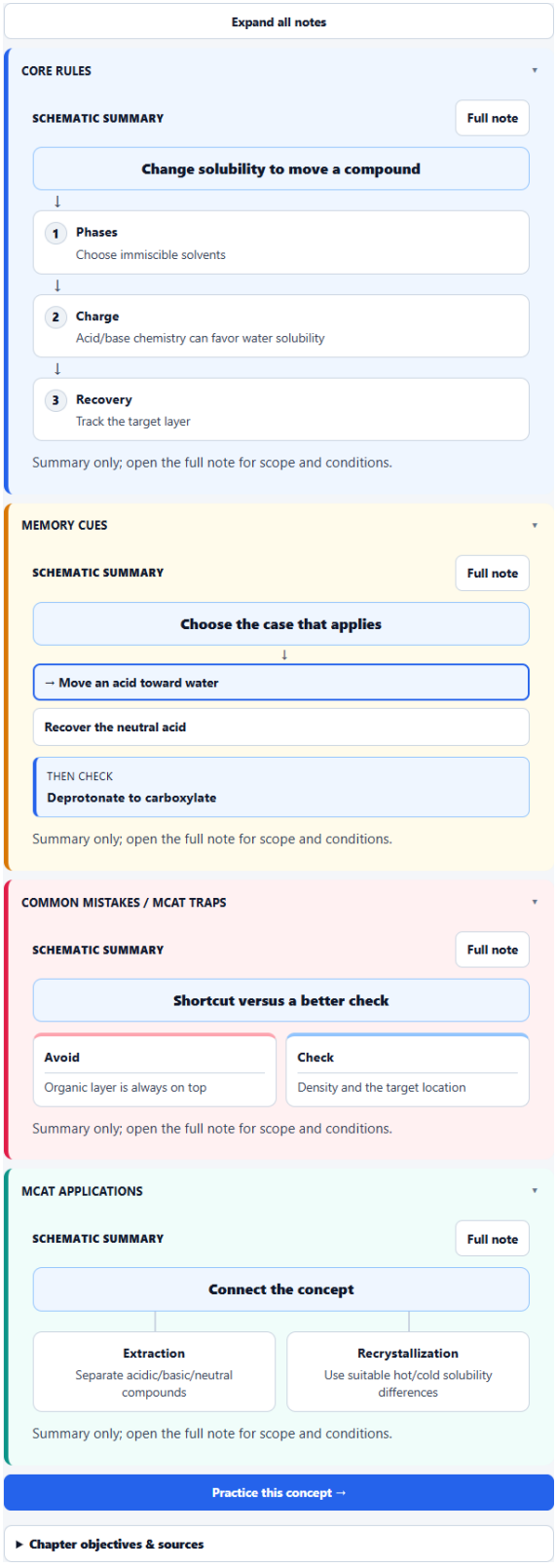
Acid/base extraction separates acidic, basic, and neutral drug-like compounds by controlling their charge and partitioning.

Schematic summary: Connect the concept

Extraction — Separate acidic/basic/neutral compounds

Recrystallization — Use suitable hot/cold solubility differences

Screenshot: solubility-visuals



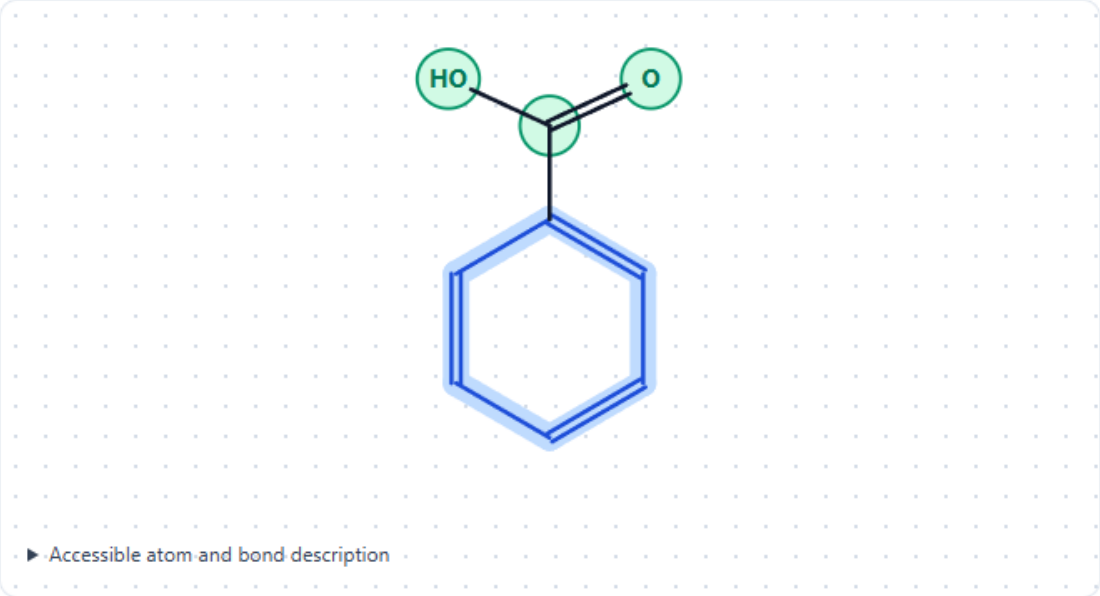
solubility:step-1 | 1. Neutral aromatic acid

Benzoic acid combines a nonpolar aromatic region with a polar acid group. Its partitioning depends on both molecular structure and ionization.

Displayed molecule ID: benzoicacid

{"highlights": {"prin": [6, 7, 8], "path": [0, 1, 2, 3, 4, 5]}, "revealName": true}

Screenshot: solubility-step-1



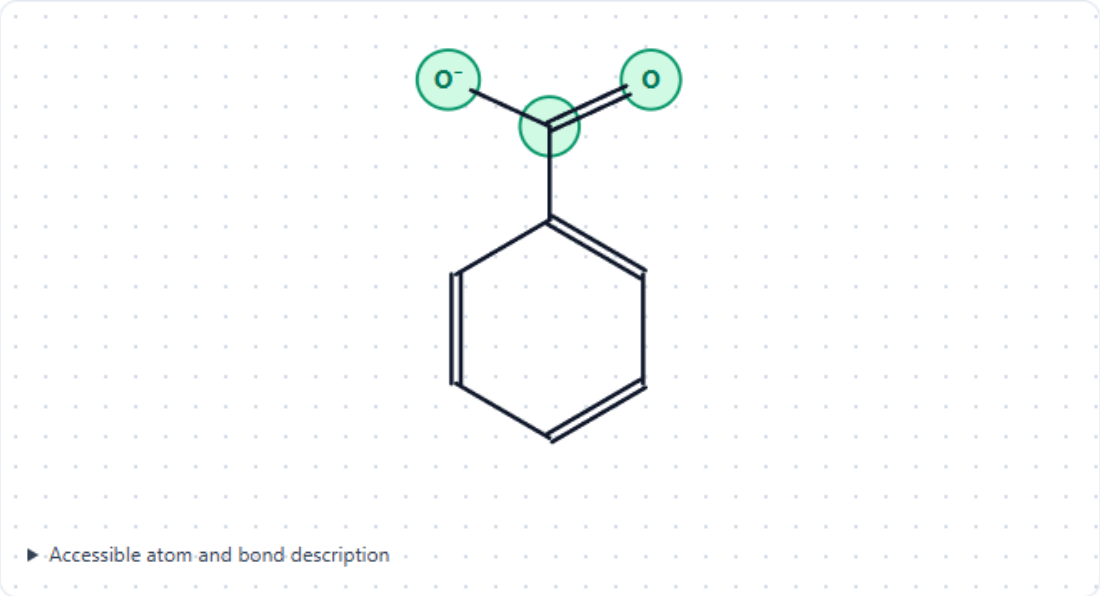
solubility:step-2 | 2. Ionize the acid

At sufficiently high pH, benzoate predominates and is generally more water soluble than neutral benzoic acid. Acidifying can regenerate neutral acid for recovery when solubility conditions permit.

Displayed molecule ID: benzoate

{"highlights": {"prin": [6, 7, 8]}, "revealName": true}

Screenshot: solubility-step-2



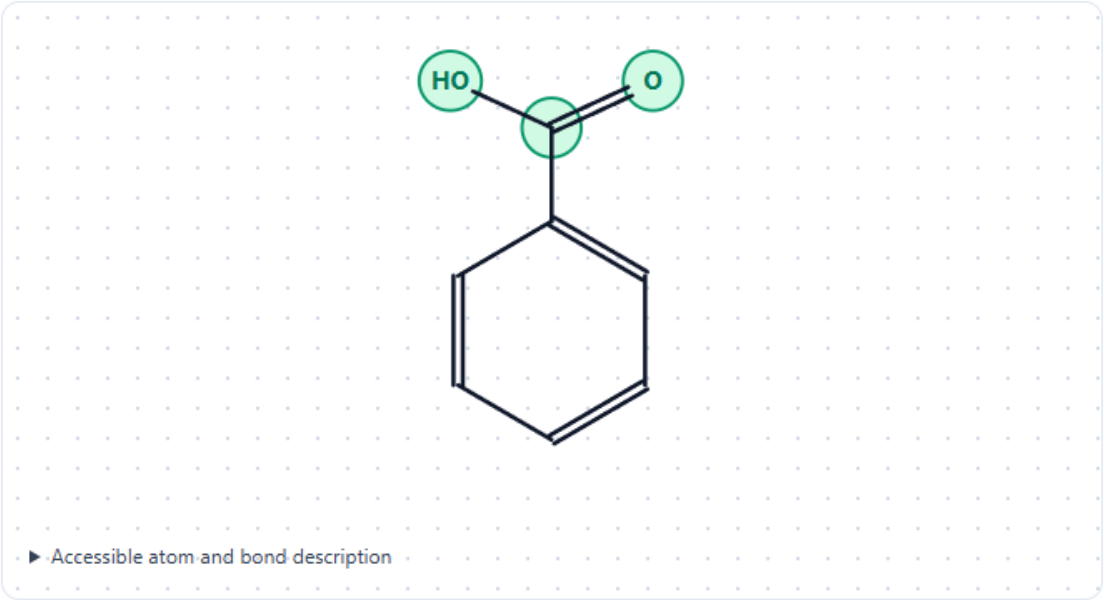
solubility:step-3 | 3. Recrystallize the recovered solid

Use a suitable solvent system in which the desired solid dissolves hot and crystallizes on cooling. Soluble impurities should remain mainly in the mother liquor; insoluble impurities can be filtered before cooling.

Displayed molecule ID: benzoicacid

{"highlights": {"prin": [6, 7, 8]}, "revealName": true}

Screenshot: solubility-step-3



distillation | Distillation and volatility

Vapor is enriched in the more volatile component

distillation:rule | Core Rules

Distillation separates liquids by volatility. Simple distillation suits sufficiently different volatilities; fractional distillation adds repeated vaporization/condensation for closer boiling mixtures. Lower pressure lowers boiling temperature, so vacuum distillation can help with high-boiling or heat-sensitive liquids. Superheating means a liquid is above its boiling temperature without sustained boiling. Nucleation sites from appropriate boiling aids or stirring can support smooth boiling; chips are not added to an already hot superheated liquid.

Schematic summary: Vapor is enriched in the more volatile component

Volatility — Select separation conditions

Close volatilities — Fractionation adds equilibrium stages

Thermal stability — Reduced pressure can lower boiling temperature

distillation:cue | Memory Cues

Fractionating column = repeated equilibrium stages. Vacuum = lower boiling temperature.

Schematic summary: Choose the case that applies

Closer volatilities — Consider fractional distillation

Decomposition risk — Consider reduced pressure

Azeotrope — Ordinary distillation can be limited

distillation:trap | Common Mistakes / MCAT Traps

Boiling-point cutoffs are heuristics, not universal rules. Azeotropes can limit ordinary distillation. Distillation does not guarantee a pure first fraction merely because one component has a lower boiling point. The supplied curriculum suggests simple distillation for boiling points roughly 25 °C or more apart and below about 150 °C, fractionation for closer boiling points, and considering vacuum for high normal boiling temperatures (around or above 150 °C) or heat-sensitive materials. These are selection heuristics, not guarantees; pressure, relative volatility, composition, apparatus and decomposition behavior matter.

Schematic summary: Shortcut versus a better check

Avoid — One fixed boiling-point gap always decides

Check — Curriculum cutoffs are heuristics; use mixture behavior and pressure

distillation:app | MCAT Applications

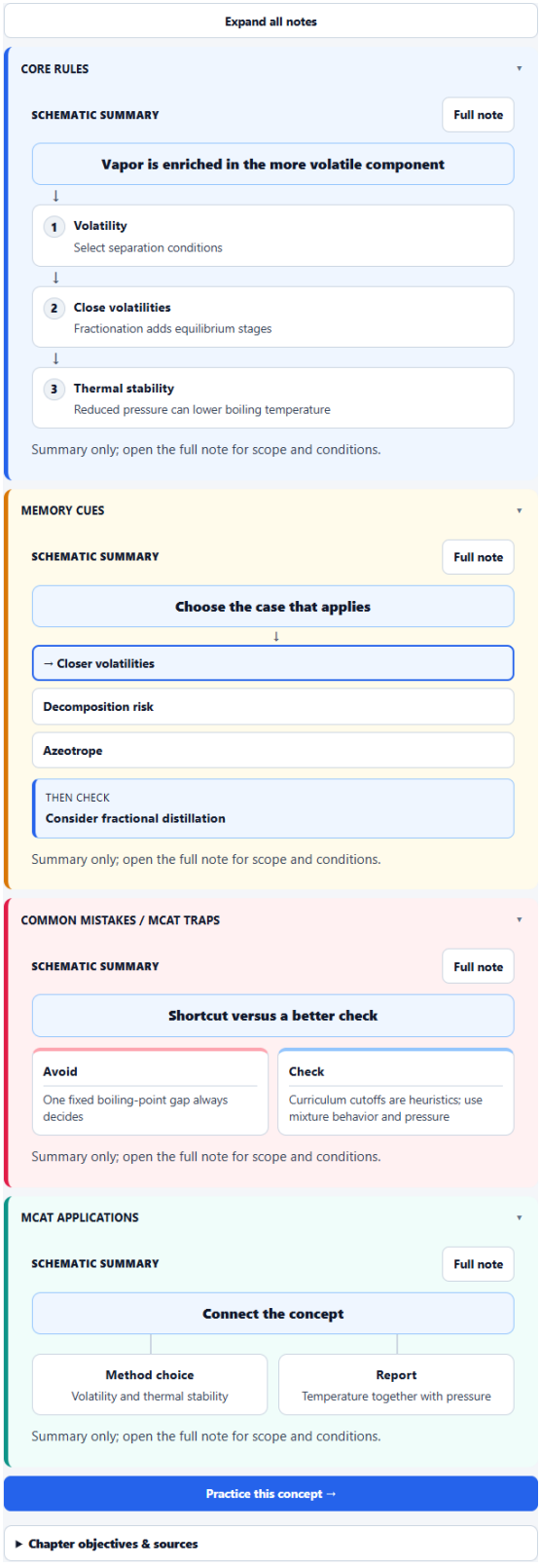
Choose a method based on volatility, thermal stability, pressure, and mixture behavior; record boiling temperatures with pressure.

Schematic summary: Connect the concept

Method choice — Volatility and thermal stability

Report — Temperature together with pressure

Screenshot: distillation-visuals



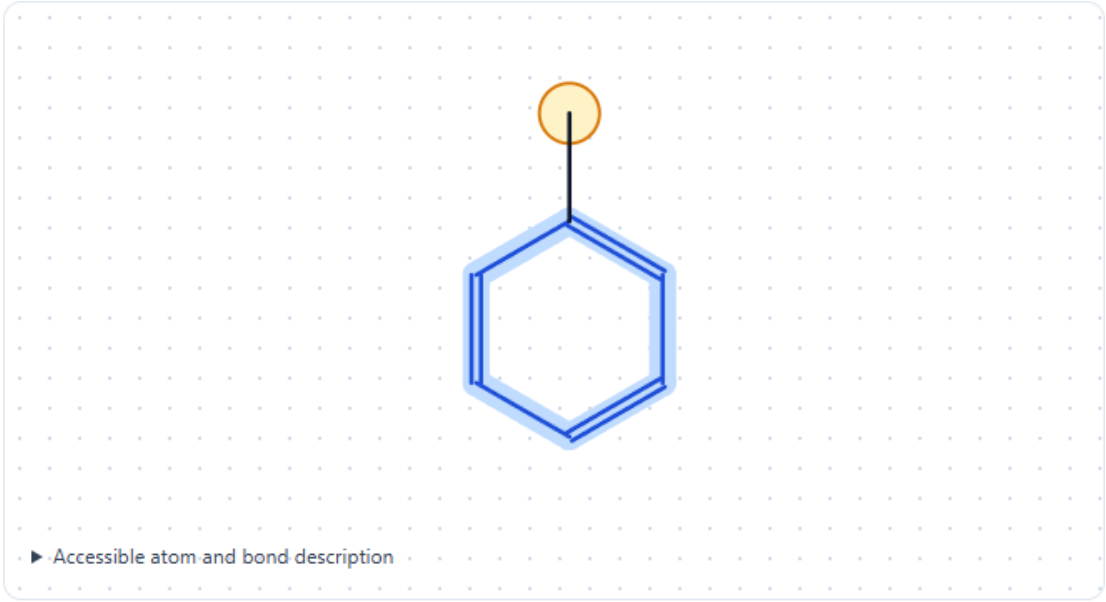
distillation:step-1 | 1. A molecular property

Toluene is a neutral volatile organic liquid. Intermolecular interactions influence its vapor pressure and boiling behavior; the structure alone does not supply an exact boiling point.

Displayed molecule ID: toluene

{"highlights": {"path": [0, 1, 2, 3, 4, 5], "sub": [6]}, "revealName": true}

Screenshot: distillation-step-1



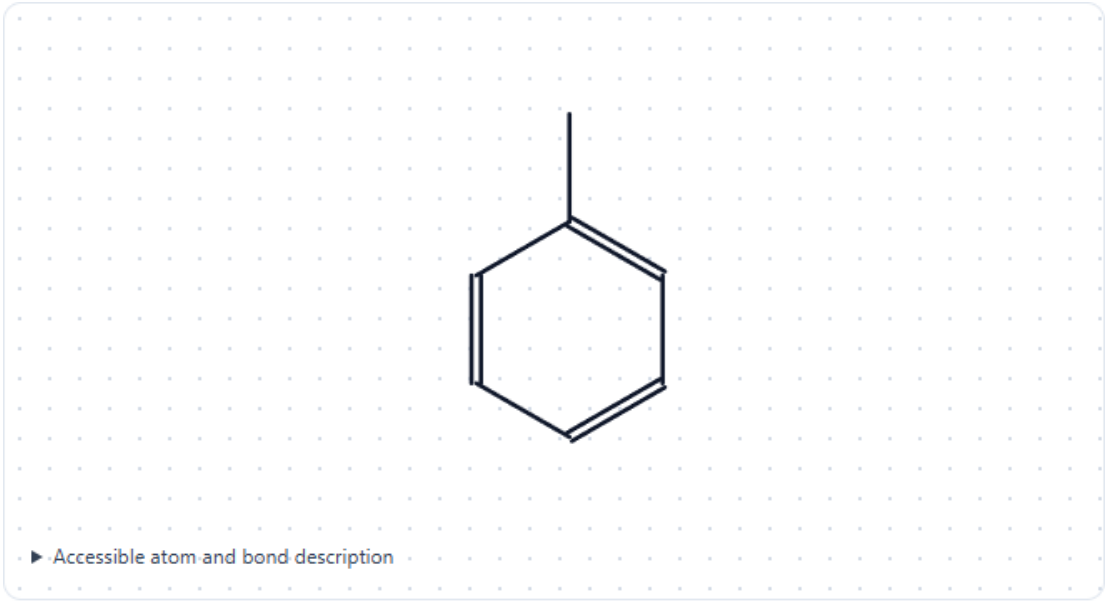
distillation:step-2 | 2. Choose a method

For a hypothetical close-boiling mixture, fractionation can improve enrichment. If the material decomposes near its atmospheric boiling point, reducing pressure may allow a lower operating temperature.

Displayed molecule ID: toluene

{"highlights": {}, "revealName": false}

Screenshot: distillation-step-2



tlc | TLC and normal-phase columns

Retention competes with the mobile phase

tlc:rule | Core Rules

In normal-phase TLC, polar silica is stationary and solvent moves upward by capillary action. Stronger attraction to silica generally lowers Rf. Rf = distance traveled by spot center / distance traveled by solvent front, both measured from the baseline. A stronger eluent usually raises Rf on silica. Paper chromatography uses polar cellulose and often an associated water layer; partitioning and adsorption depend on the solvent system. Invisible TLC spots can be visualized by UV or suitable iodine, vanillin or phosphomolybdic-acid stains; detection depends on the analyte.

Schematic summary: Retention competes with the mobile phase

Baseline — Common origin for measurements

Spot-center distance — Rf numerator

Solvent-front distance — Rf denominator; ratio has no units

tlc:cue | Memory Cues

Stronger stationary-phase binding → slower travel. Rf has no units.

Schematic summary: Choose the case that applies

Normal-phase silica — Stronger silica binding generally lowers Rf

Reversed phase — Nonpolar stationary phase changes polarity trends

Comparing Rf — Use matching plate and solvent conditions

tlc:trap | Common Mistakes / MCAT Traps

Rf depends on plate, solvent, and conditions; it is not a permanent molecular constant. Equal Rf is supporting evidence, not definitive identification. In reversed-phase systems the stationary phase is nonpolar, so polarity trends differ.

Schematic summary: Shortcut versus a better check

Avoid — Equal Rf proves identity or absolute purity

Check — Supporting evidence, not a permanent molecular constant

tlc:app | MCAT Applications

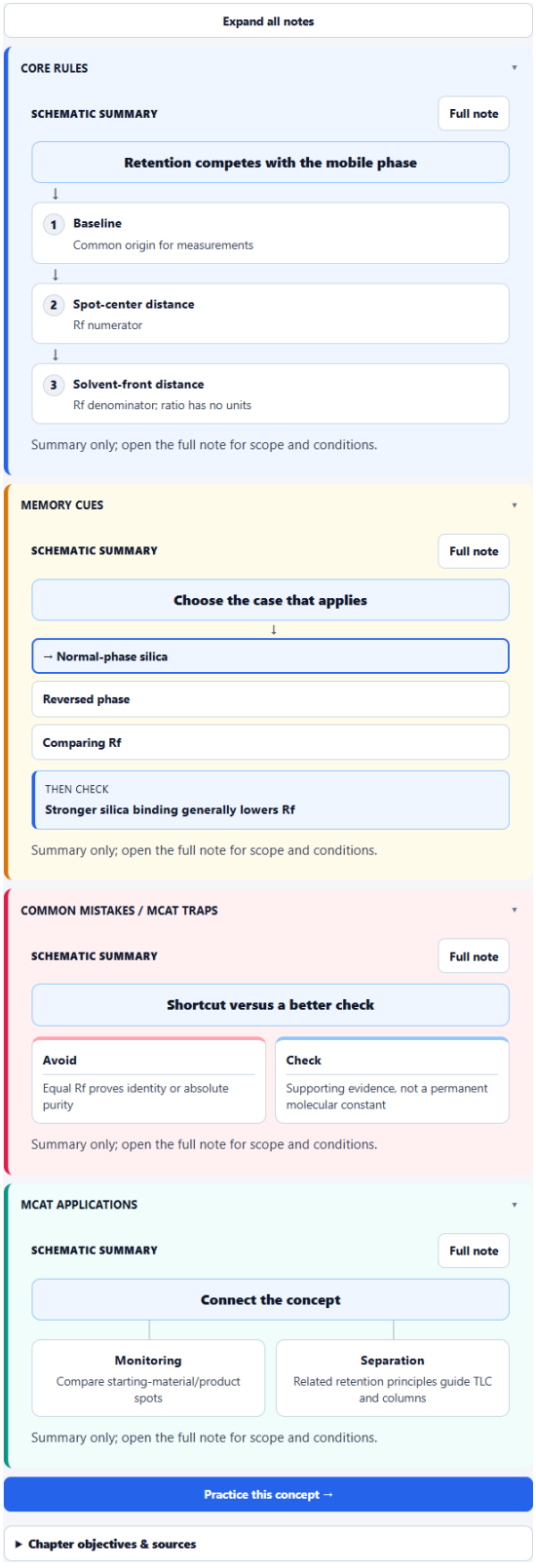
TLC can follow reaction progress by comparing starting-material and product spots under the same conditions; a single spot is not proof of absolute purity. Preparative TLC can isolate material from bands; flash column chromatography uses pressure to drive mobile phase through a packed stationary phase. The analytical TLC trend is not an experimental Rf prediction.

Schematic summary: Connect the concept

Monitoring — Compare starting-material/product spots

Separation — Related retention principles guide TLC and columns

Screenshot: tlc-visuals



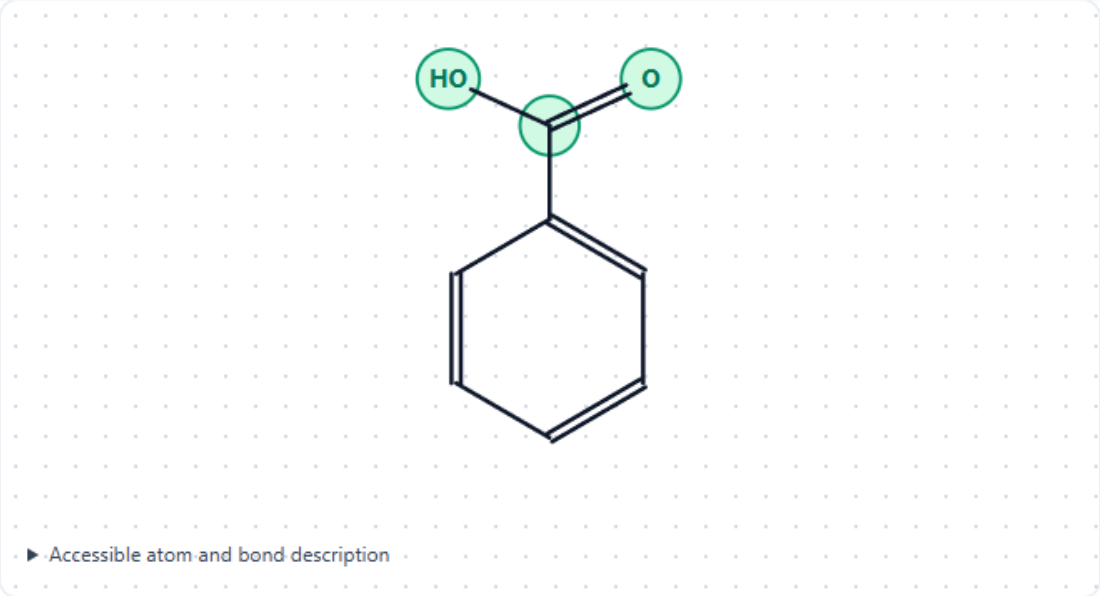
tlc:step-1 | 1. Identify polar interactions

The carboxyl group can interact strongly with silica. Actual retention depends on the eluent, ionization, and other experimental details.

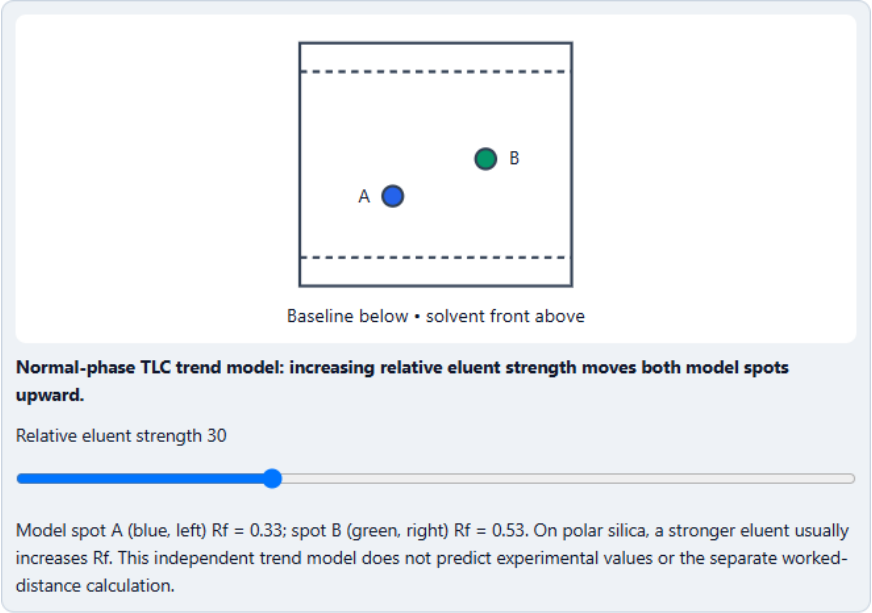
Displayed molecule ID: benzoicacid

{"highlights": {"prin": [6, 7, 8]}, "revealName": true}

Screenshot: tlc-step-1



Screenshot: tlc-step-1-diagram



{"diagramSpecification": {"type": "tlc", "caption": "Normal-phase TLC trend model: increasing relative eluent strength moves both model spots upward."}}

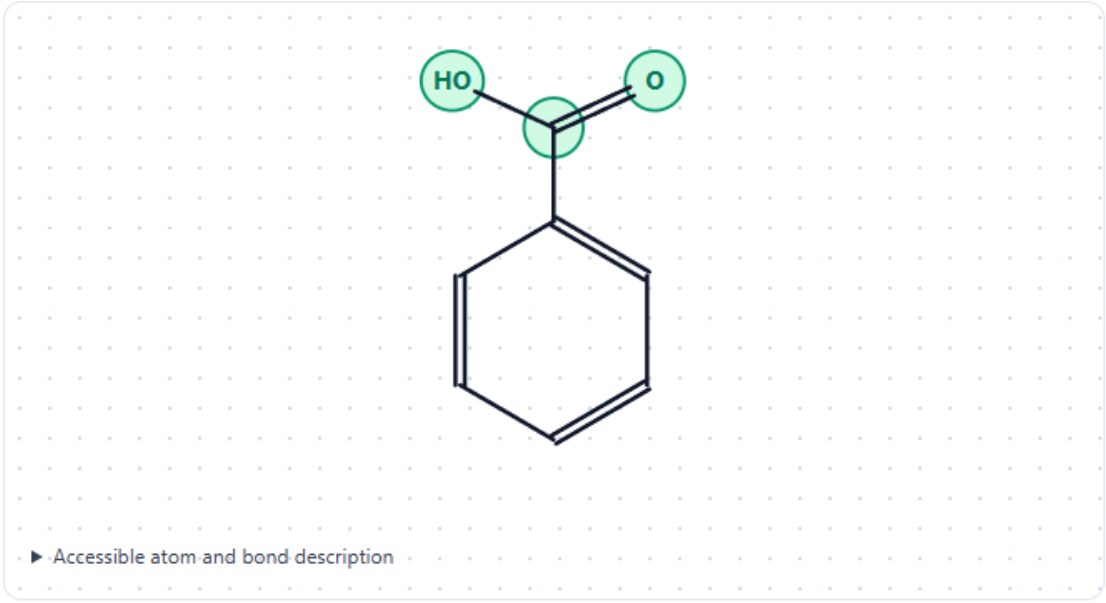
tlc:step-2 | 2. Calculate normalized travel

A spot traveling 1.8 cm while the front travels 6.0 cm has Rf=0.30. Use the same origin for both measurements. The schematic slider is a trend illustration, not a prediction for this molecule.

Displayed molecule ID: benzoicacid

{"highlights": {"prin": [6, 7, 8]}, "revealName": false}

Screenshot: tlc-step-2



specialized | Choose a chromatography method

Match the separation to a molecular property

specialized:rule | Core Rules

Ion exchange separates by charge: a positively charged anion exchanger binds anions; a negatively charged cation exchanger binds cations. Size exclusion separates by access to pores, with larger species generally eluting first. Affinity chromatography uses specific binding. GC requires adequate volatility and thermal stability; HPLC uses a liquid mobile phase under pressure and can handle many nonvolatile analytes.

Schematic summary: Match the separation to a molecular property

Charge — Ion exchange

Pore access — Size exclusion

Specific interaction — Affinity

specialized:cue | Memory Cues

Ion exchange: charge. Size exclusion: pore access. Affinity: specific interaction.

Schematic summary: Choose the case that applies

Anions — Bind a positively charged anion exchanger

Cations — Bind a negatively charged cation exchanger

Size exclusion — Larger species generally elute first

specialized:trap | Common Mistakes / MCAT Traps

Small molecules enter more pore volume and usually elute later in size exclusion. “Anion exchanger” names the species it binds, not the resin’s own charge. Reversed-phase HPLC usually retains hydrophobic analytes more strongly than polar ones under comparable conditions.

Schematic summary: Shortcut versus a better check

Avoid — Anion exchanger must be negative

Check — The name refers to the ions it binds

specialized:app | MCAT Applications

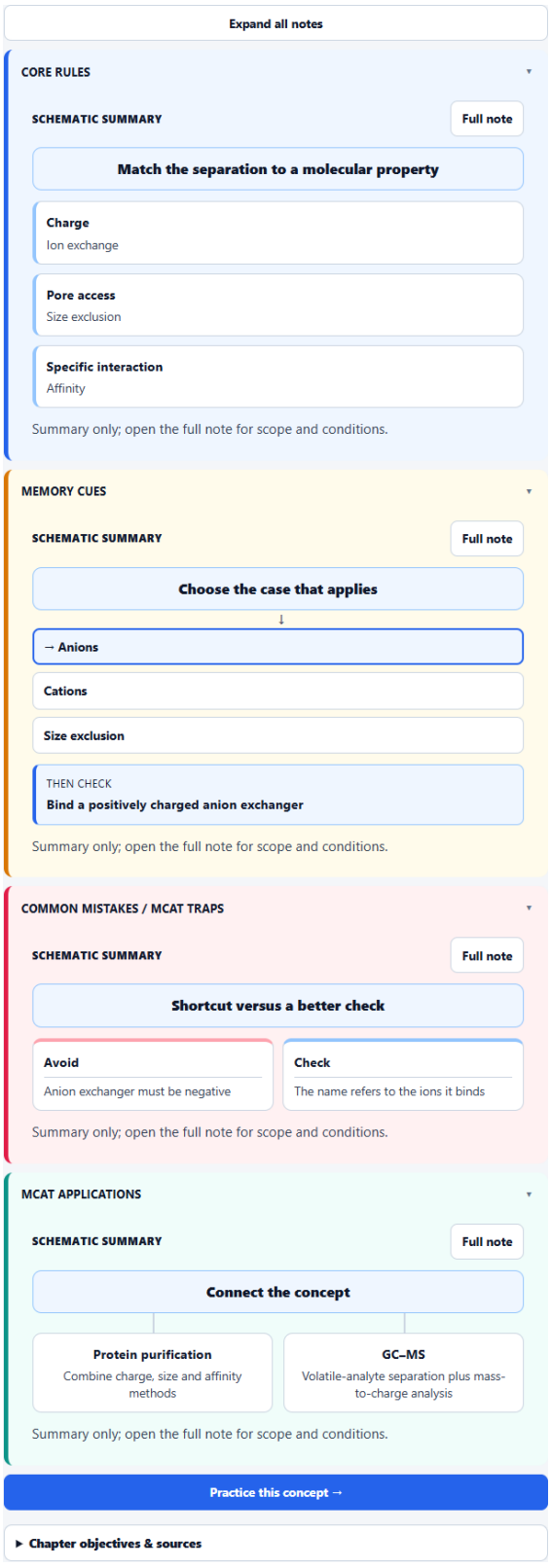
Protein purification often combines orthogonal methods: charge, size, and affinity. Choose based on protein stability as well as separation power. GC–MS couples separation of suitably volatile analytes with ionization and mass-to-charge analysis. Fragmentation depends on ionization mode; molecular-ion and fragment patterns can support identification, but not every peak is the intact molecular mass.

Schematic summary: Connect the concept

Protein purification — Combine charge, size and affinity methods

GC–MS — Volatile-analyte separation plus mass-to-charge analysis

Screenshot: specialized-visuals



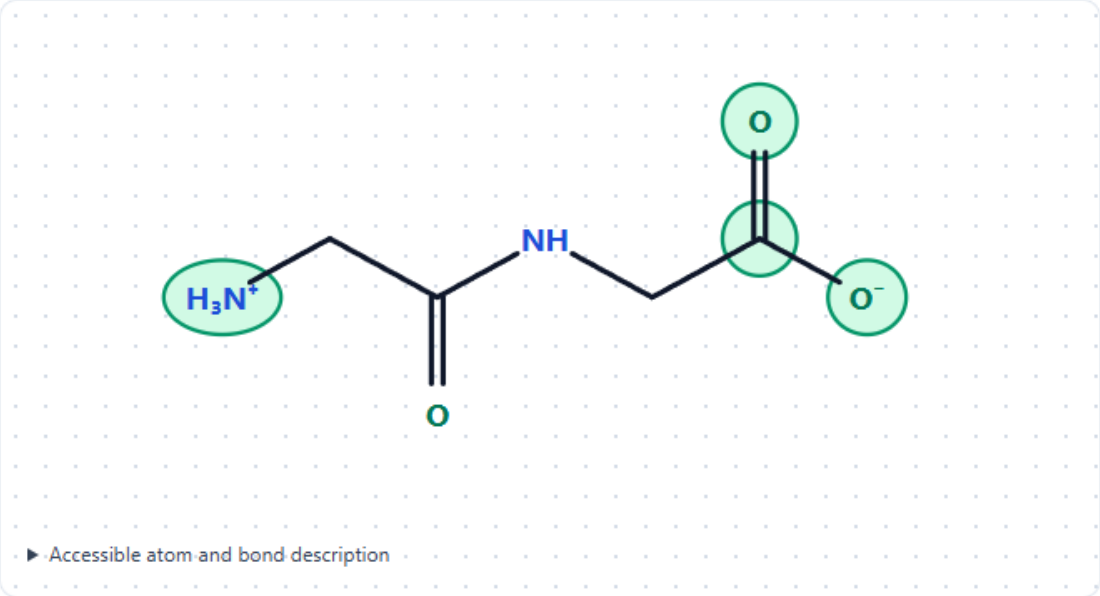
specialized:step-1 | 1. Charge depends on conditions

This peptide has ionizable termini. At pH above its pI it tends toward negative net charge, which can favor binding to a positive anion-exchange resin. Real proteins may have many additional ionizable side chains.

Displayed molecule ID: glycylglycine

{"highlights": {"prin": [0, 6, 7, 8]}, "revealName": true}

Screenshot: specialized-step-1



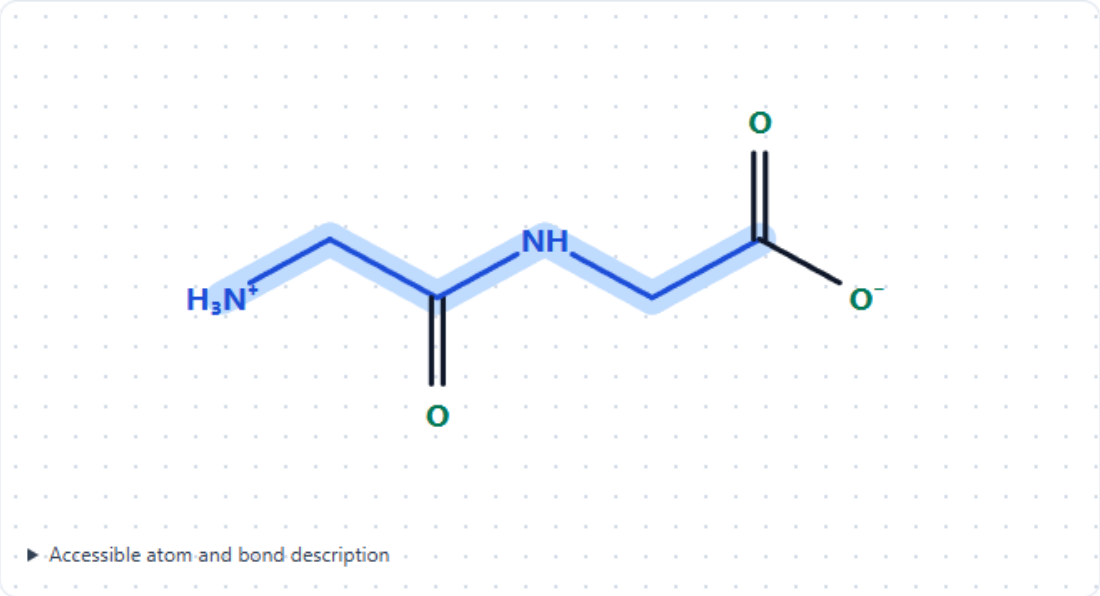
specialized:step-2 | 2. Choose by analyte behavior

An ionic nonvolatile peptide is usually better suited to a liquid-phase method than direct GC without derivatization. Larger proteins can be separated by hydrodynamic size using porous beads; mass alone is not the only determinant.

Displayed molecule ID: glycylglycine

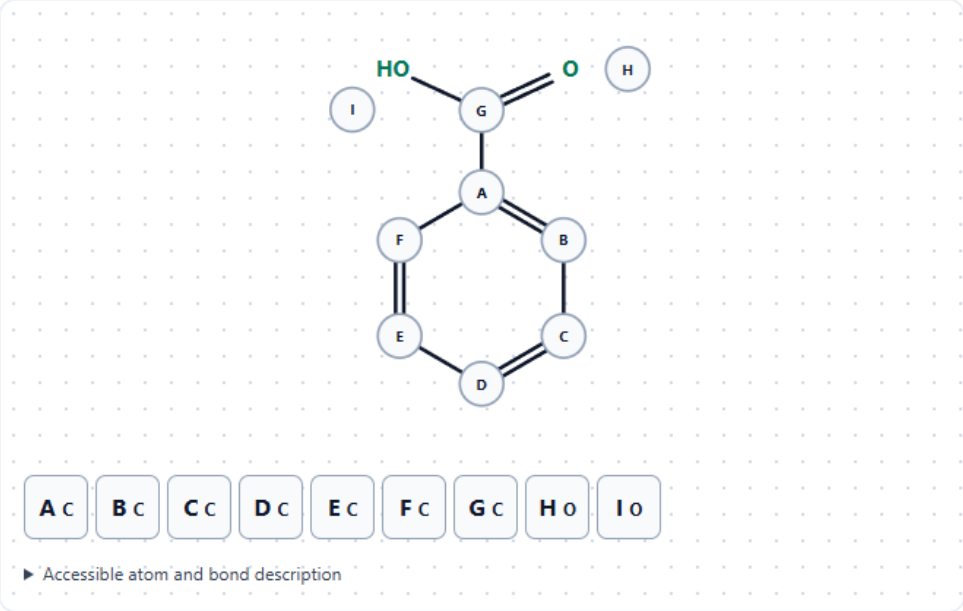
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Screenshot: specialized-step-2



acid-extraction | Plan an acid–base separation

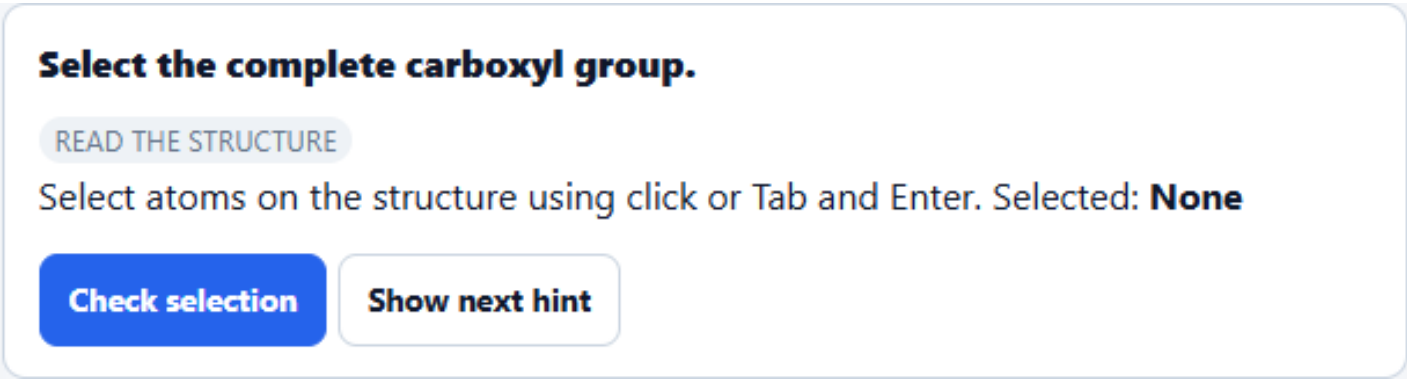
Screenshot: acid-extraction-stage-1-molecule



acid-extraction:stage-1 | Select the complete carboxyl group.

Answer type: atoms

Screenshot: acid-extraction-stage-1-question



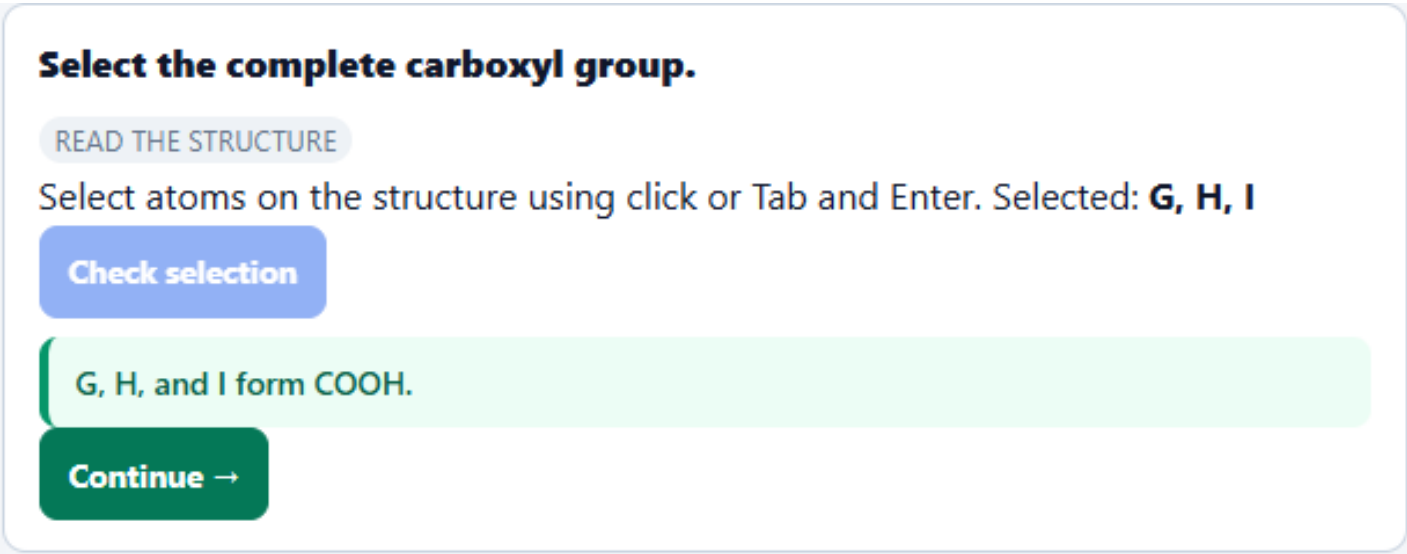
{"authorAnswer": [6, 7, 8], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [6, 7, 8]}}

Hint 1: Find the carbon outside the ring bonded to two O atoms.

Hint 2: Include the carbon and both oxygens.

Explanation: G, H, and I form COOH.

Screenshot: acid-extraction-stage-1-feedback



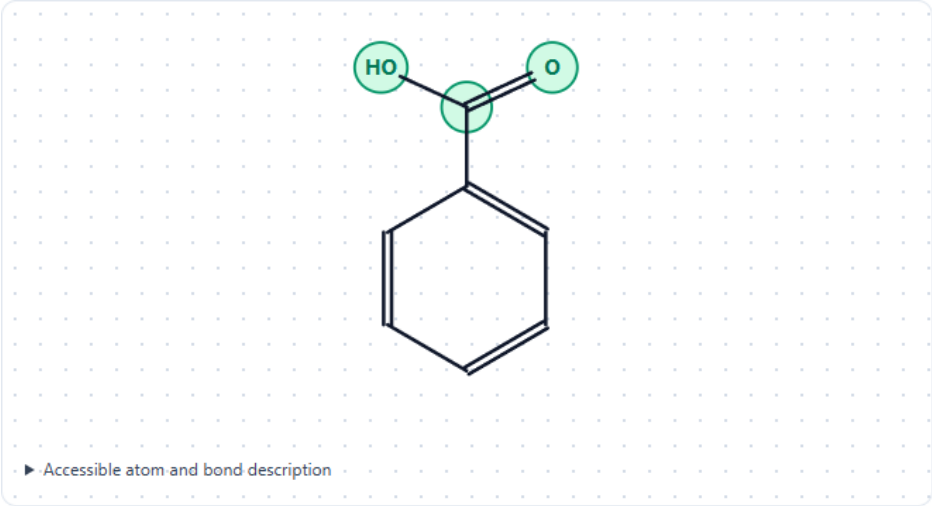
acid-extraction:stage-2 | To favor transfer of this acid into an aqueous phase, choose...

Answer type: choice

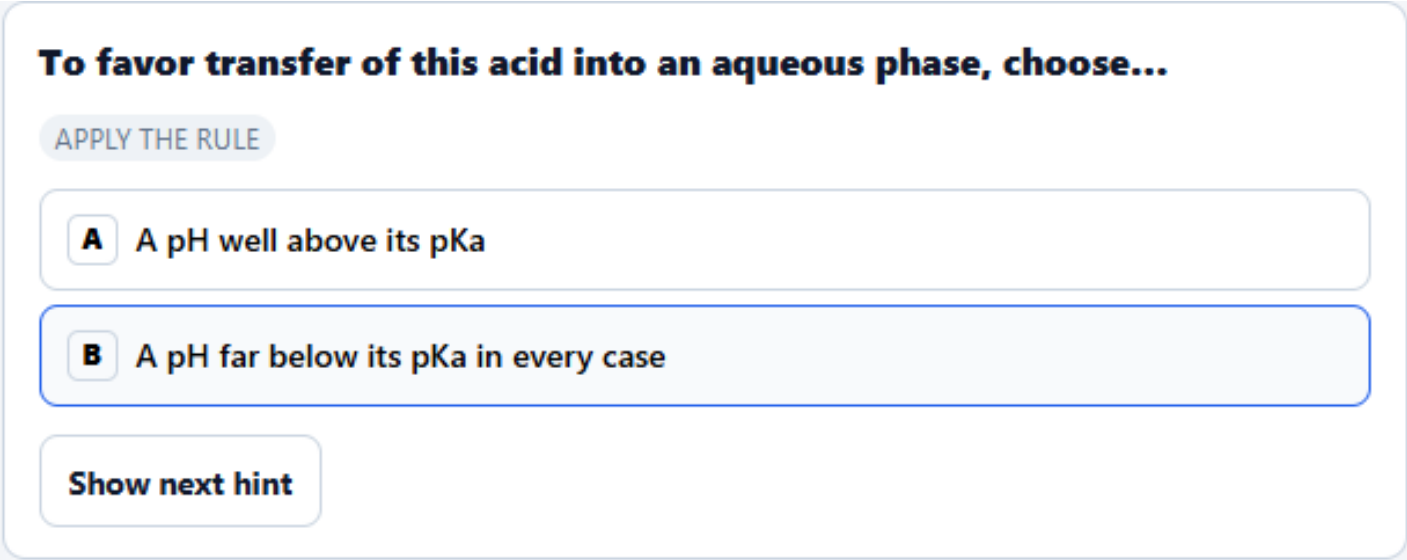
Option 0 (A): A pH well above its pKa

Option 1 (B): A pH far below its pKa in every case

Screenshot: acid-extraction-stage-2-molecule



Screenshot: acid-extraction-stage-2-question



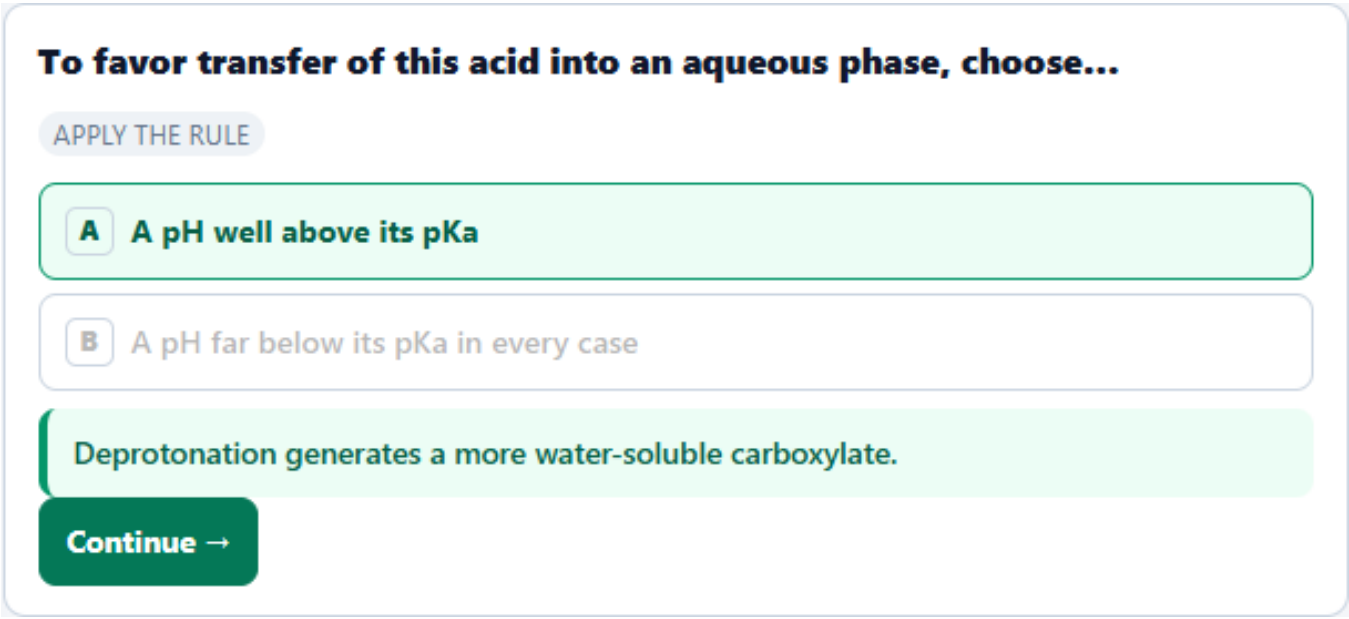
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use ionization to change partitioning.

Hint 2: High pH removes the acidic proton.

Explanation: Deprotonation generates a more water-soluble carboxylate.

Screenshot: acid-extraction-stage-2-feedback



acid-extraction:stage-3 | To collect recrystallized solid from cold mother liquor, use...

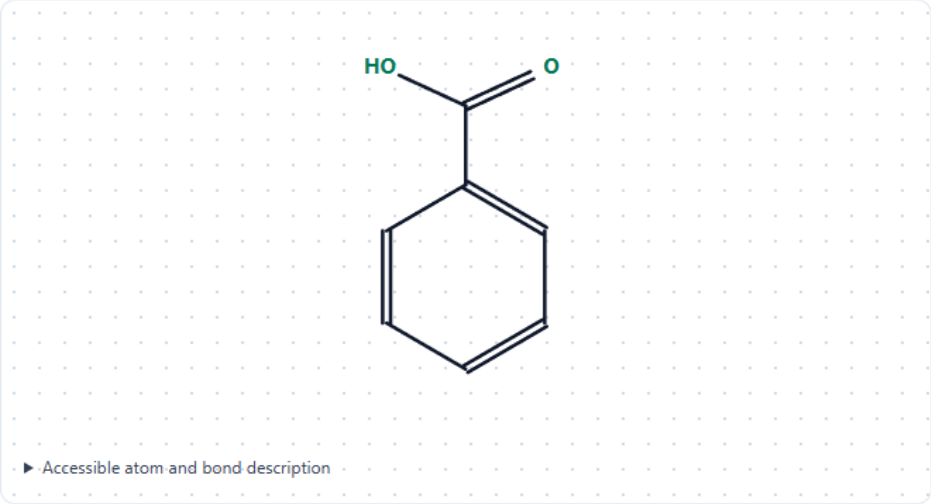
Answer type: choice

Option 0 (A): Vacuum filtration

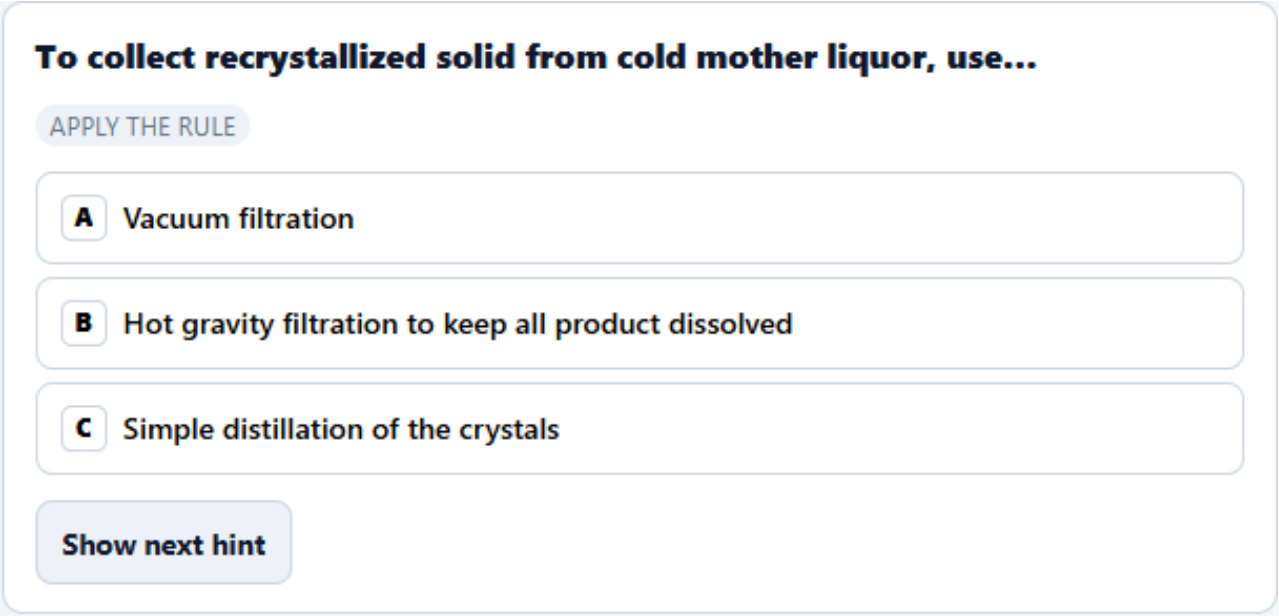
Option 1 (B): Hot gravity filtration to keep all product dissolved

Option 2 (C): Simple distillation of the crystals

Screenshot: acid-extraction-stage-3-molecule



Screenshot: acid-extraction-stage-3-question



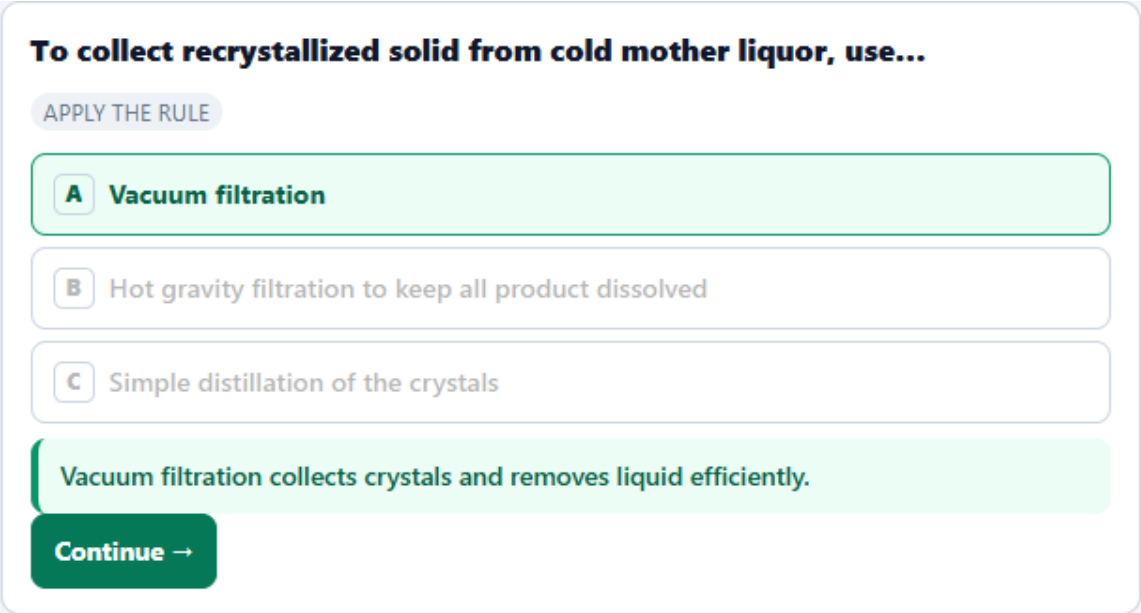
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Identify whether the target is solid or dissolved.

Hint 2: The crystals are the desired residue.

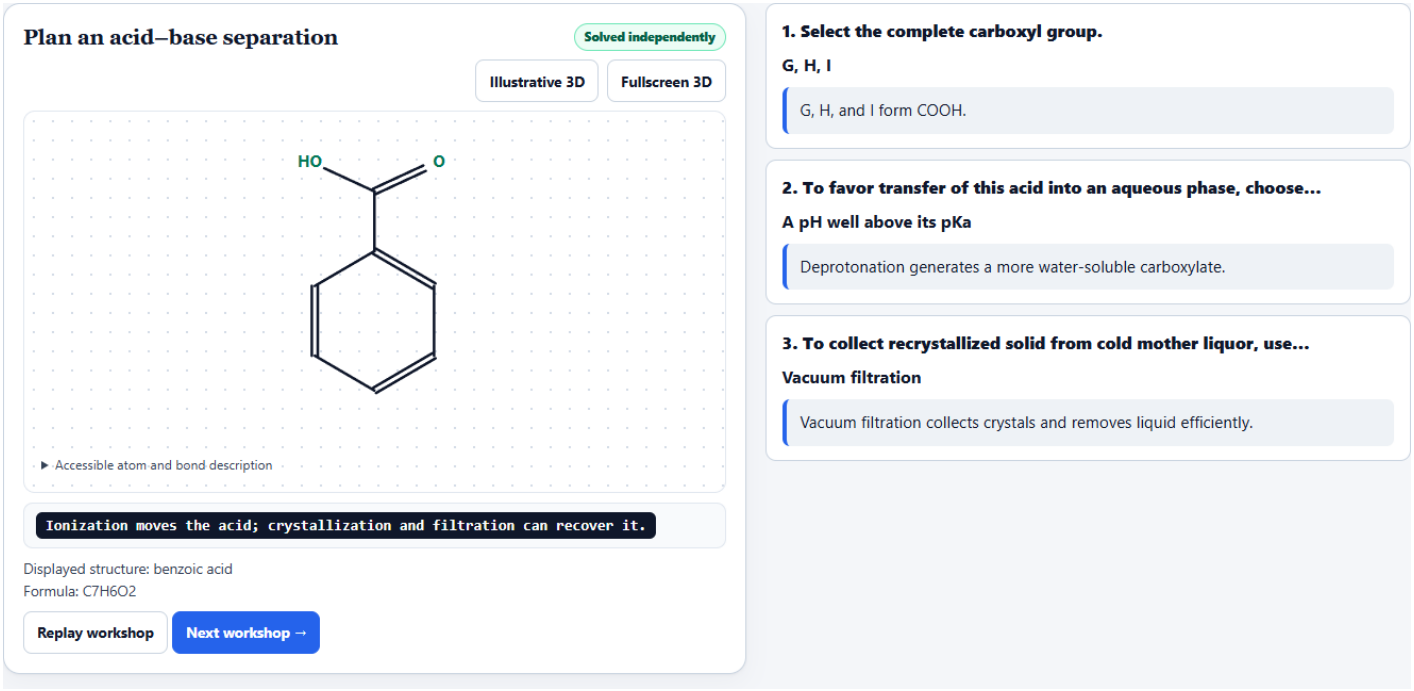
Explanation: Vacuum filtration collects crystals and removes liquid efficiently.

Screenshot: acid-extraction-stage-3-feedback



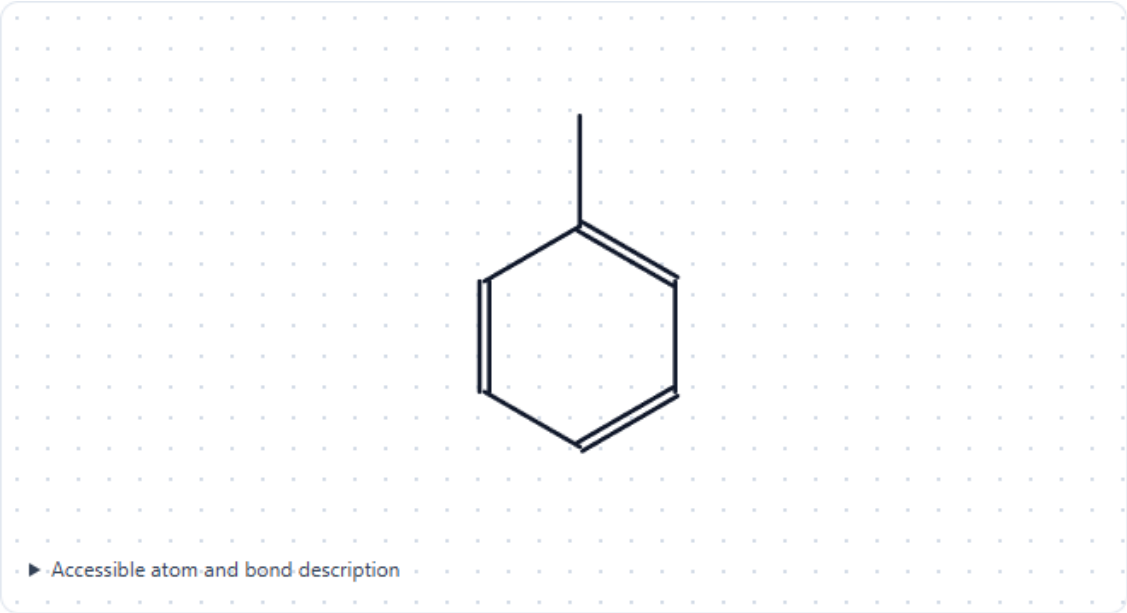
Completion summary: Ionization moves the acid; crystallization and filtration can recover it.

Screenshot: acid-extraction-complete



choose-distillation | Choose the separation conditions

Screenshot: choose-distillation-stage-1-molecule



choose-distillation:stage-1 | A close-boiling liquid mixture is generally better separated with...

Answer type: choice

Option 0 (A): Fractional distillation

Option 1 (B): Simple filtration

Screenshot: choose-distillation-stage-1-question



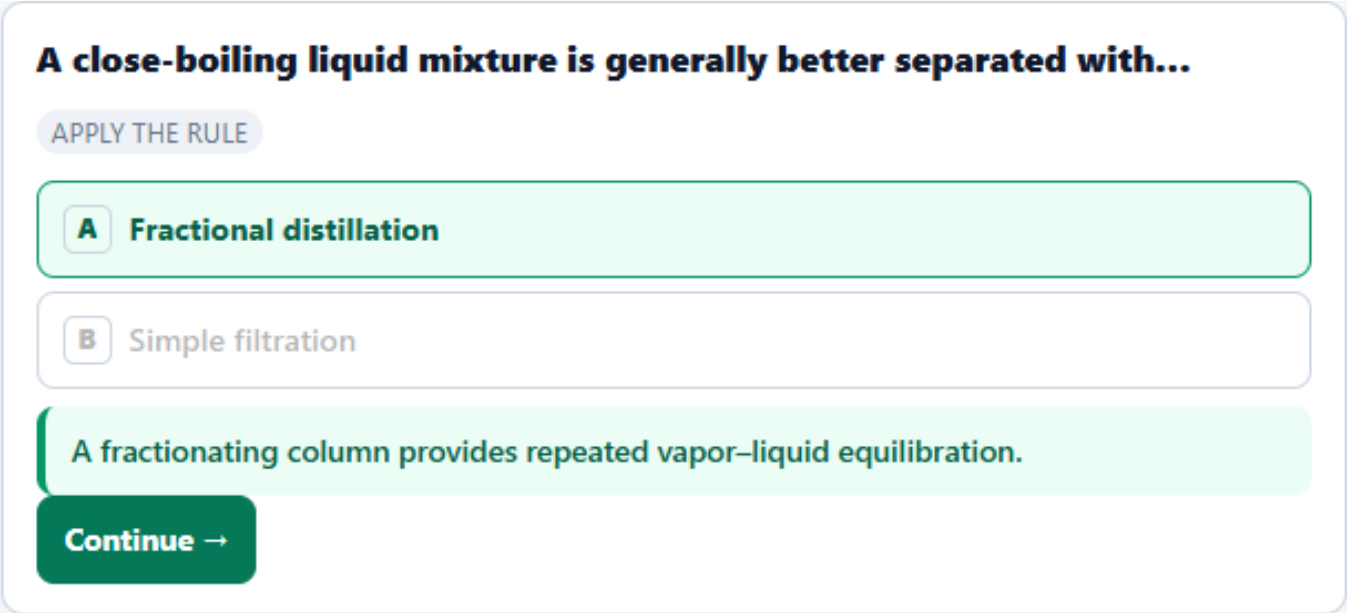
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Both components are liquids.

Hint 2: Repeated enrichment helps when volatility differences are small.

Explanation: A fractionating column provides repeated vapor–liquid equilibration.

Screenshot: choose-distillation-stage-1-feedback



choose-distillation:stage-2 | Reducing external pressure generally makes the boiling temperature...

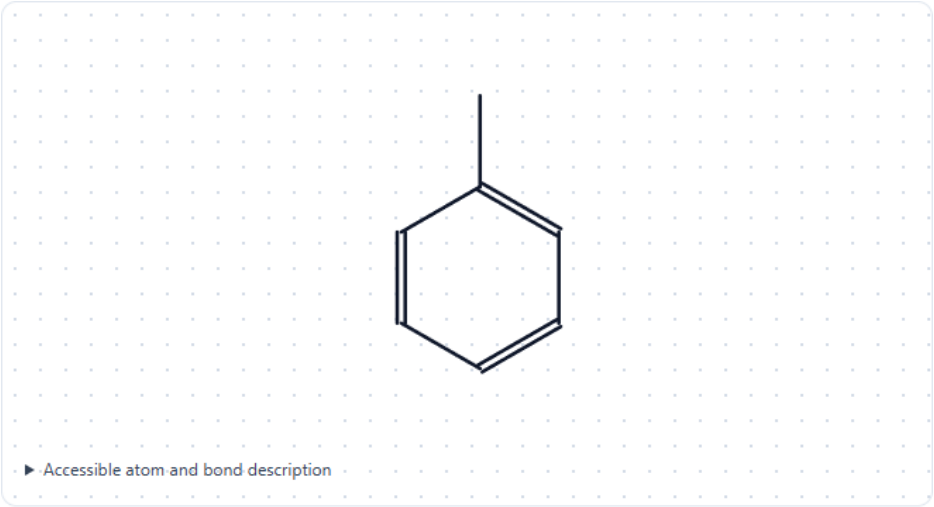
Answer type: choice

Option 0 (A): Lower

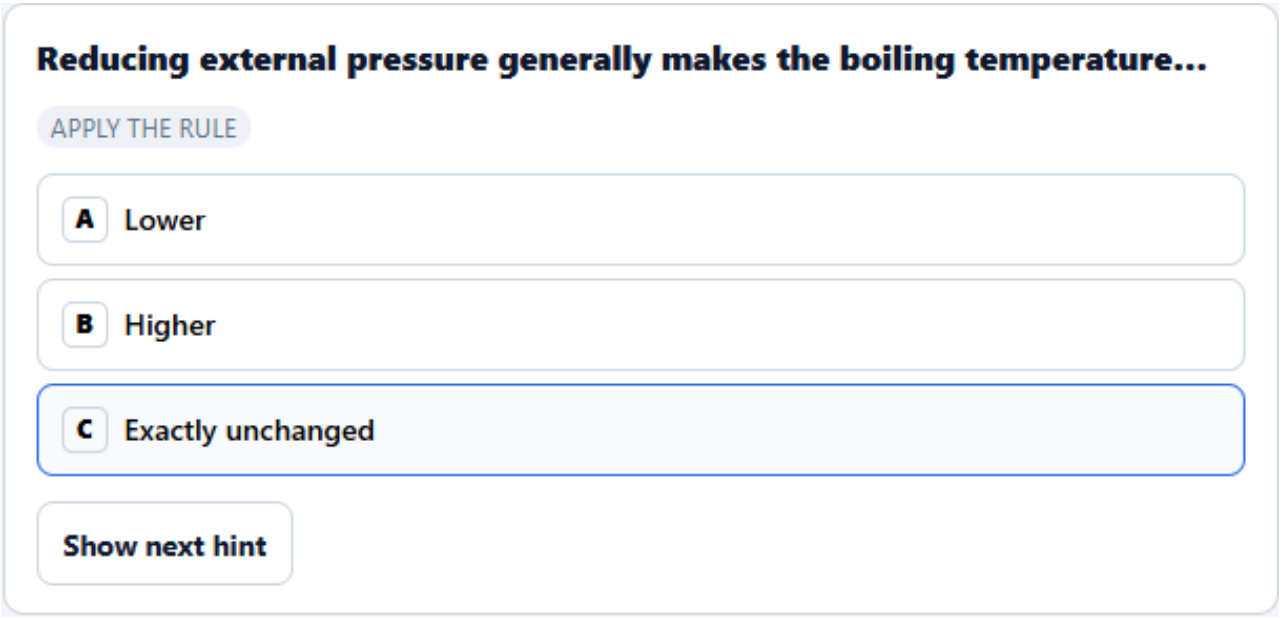
Option 1 (B): Higher

Option 2 (C): Exactly unchanged

Screenshot: choose-distillation-stage-2-molecule



Screenshot: choose-distillation-stage-2-question



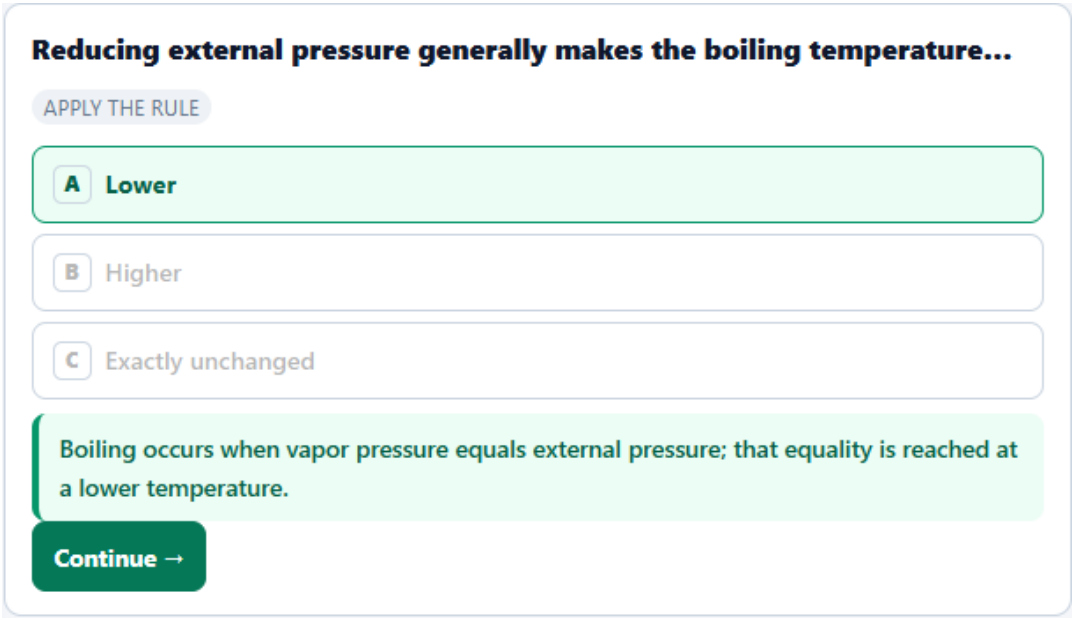
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the condition for boiling.

Hint 2: Less external pressure requires less vapor pressure.

Explanation: Boiling occurs when vapor pressure equals external pressure; that equality is reached at a lower temperature.

Screenshot: choose-distillation-stage-2-feedback



choose-distillation:stage-3 | Which can limit ordinary fractional distillation?

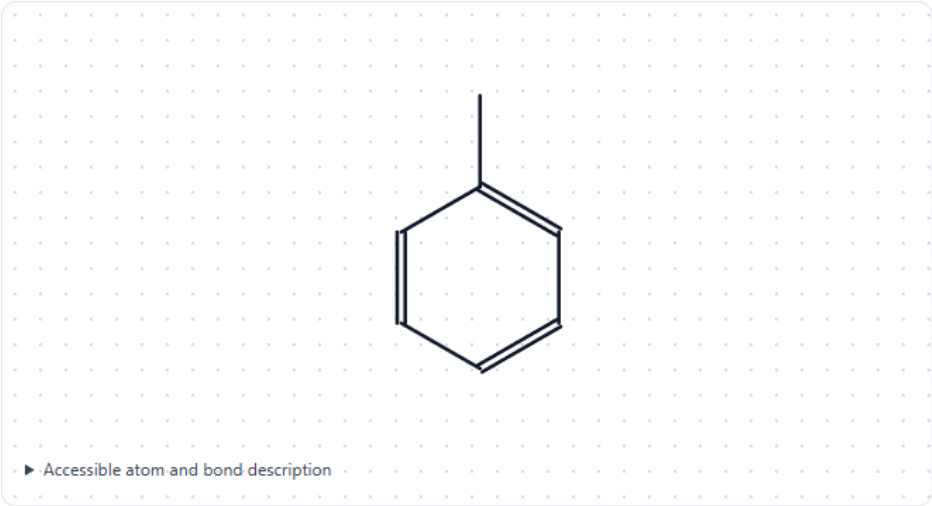
Answer type: choice

Option 0 (A): An azeotrope

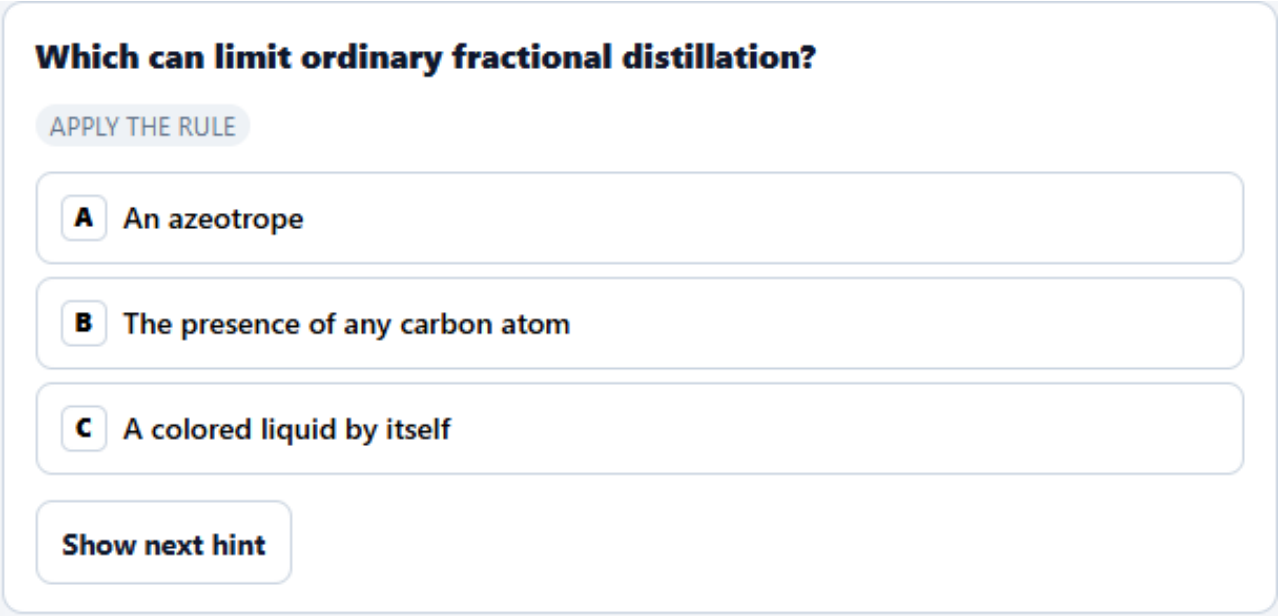
Option 1 (B): The presence of any carbon atom

Option 2 (C): A colored liquid by itself

Screenshot: choose-distillation-stage-3-molecule



Screenshot: choose-distillation-stage-3-question



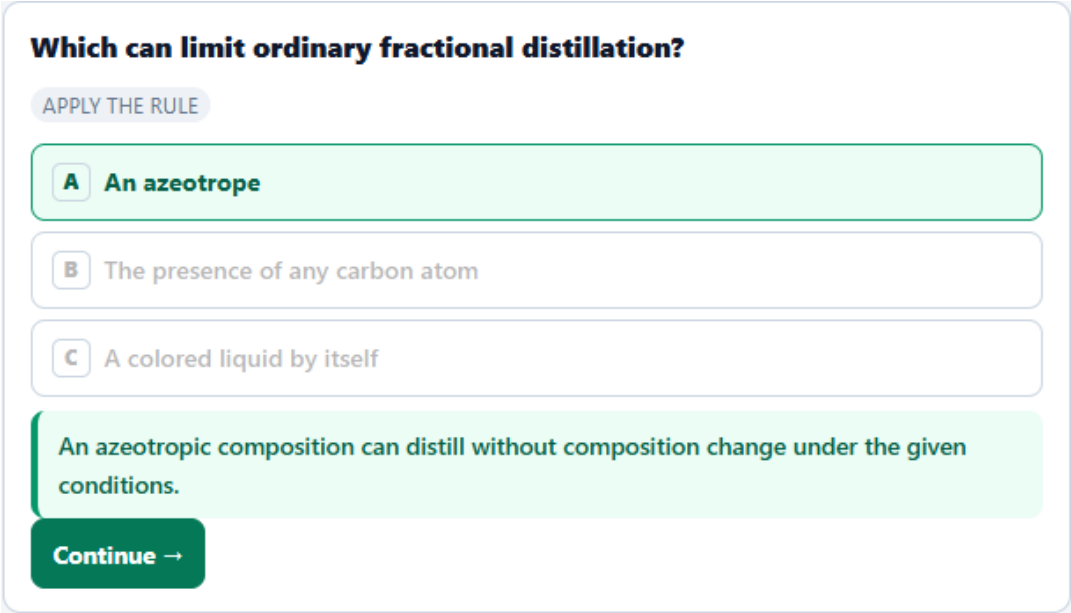
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider nonideal mixture behavior.

Hint 2: A column cannot always surpass the equilibrium limit.

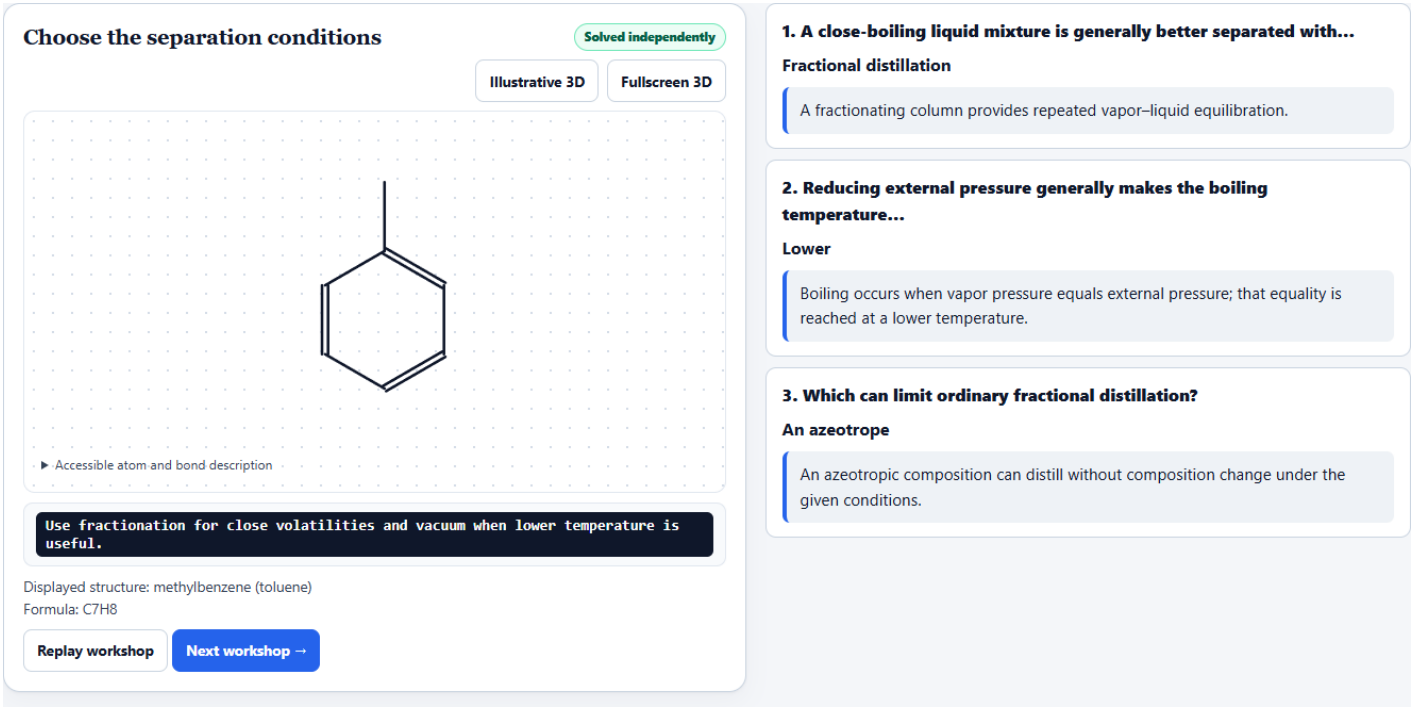
Explanation: An azeotropic composition can distill without composition change under the given conditions.

Screenshot: choose-distillation-stage-3-feedback



Completion summary: Use fractionation for close volatilities and vacuum when lower temperature is useful.

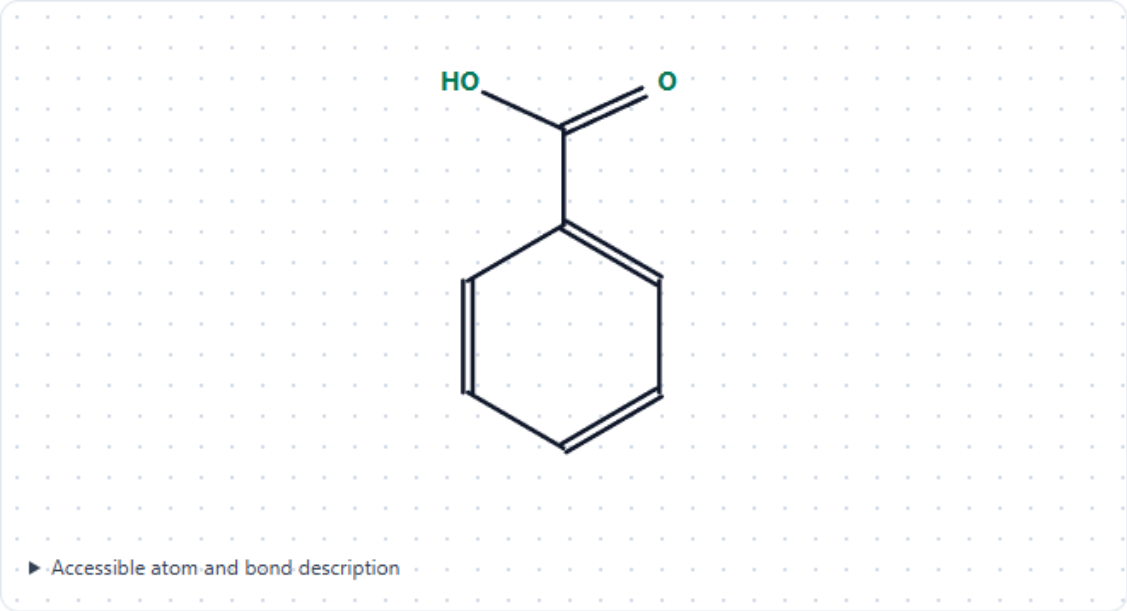
Screenshot: choose-distillation-complete



rf-calculation | Calculate and interpret Rf

Use only the stated travel distances for this calculation. The separate Learn & Explore slider illustrates an independent TLC trend, not these measurements.

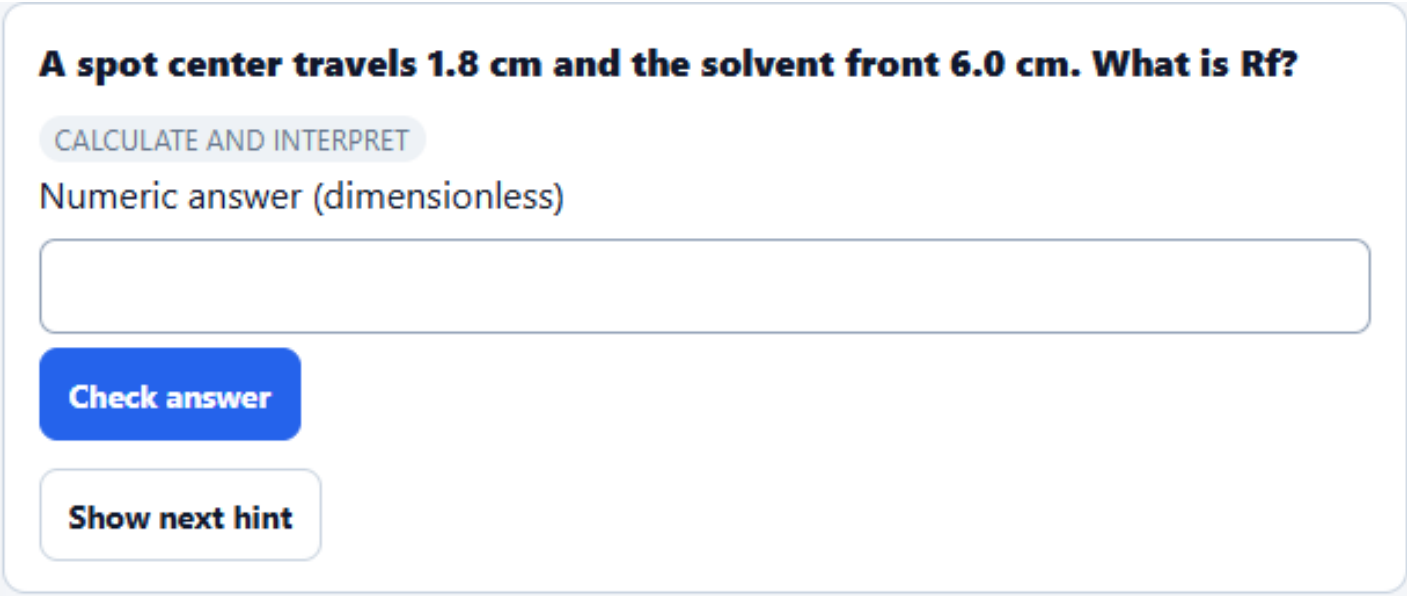
Screenshot: rf-calculation-stage-1-molecule



rf-calculation:stage-1 | A spot center travels 1.8 cm and the solvent front 6.0 cm. What is Rf?

Answer type: number | unit: dimensionless

Screenshot: rf-calculation-stage-1-question



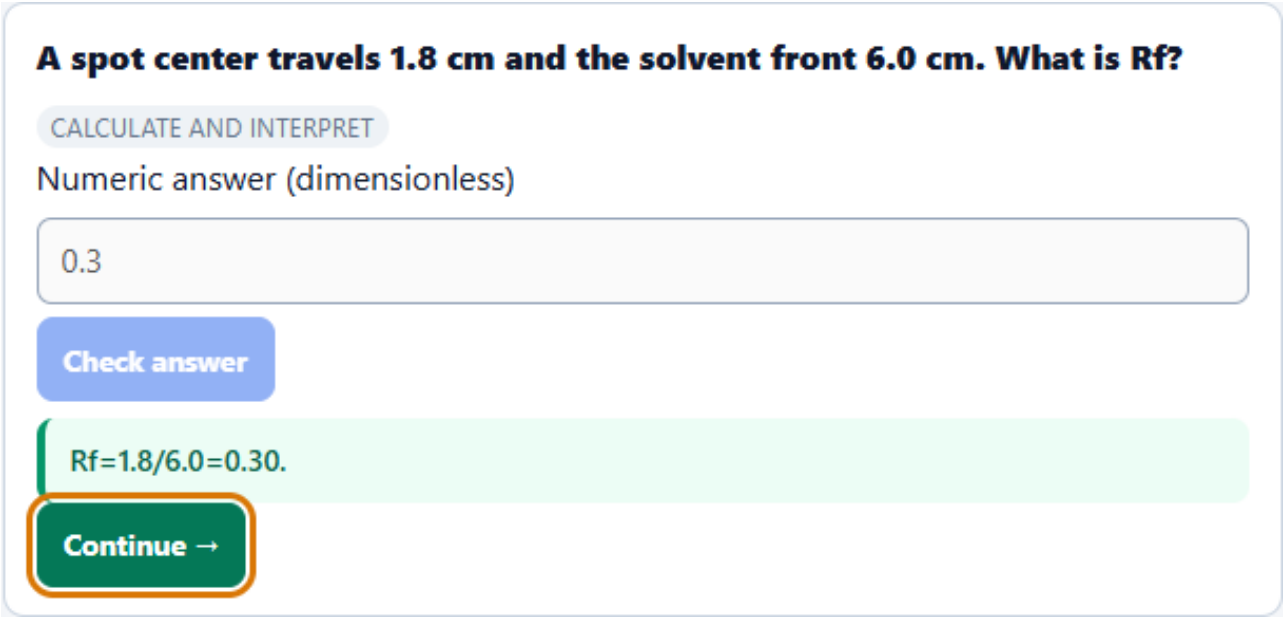
{"authorAnswer": 0.3, "tolerance": 0.005, "highlightsAfterCorrectAnswer": {}}

Hint 1: Divide spot distance by front distance.

Hint 2: Both distances start at the baseline.

Explanation: Rf=1.8/6.0=0.30.

Screenshot: rf-calculation-stage-1-feedback



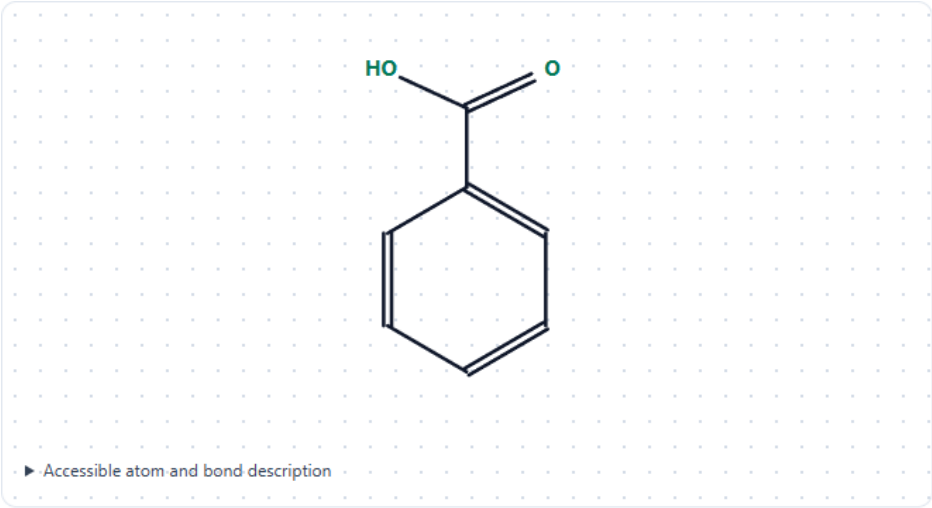
rf-calculation:stage-2 | On normal-phase silica, a stronger eluent usually makes Rf…

Answer type: choice

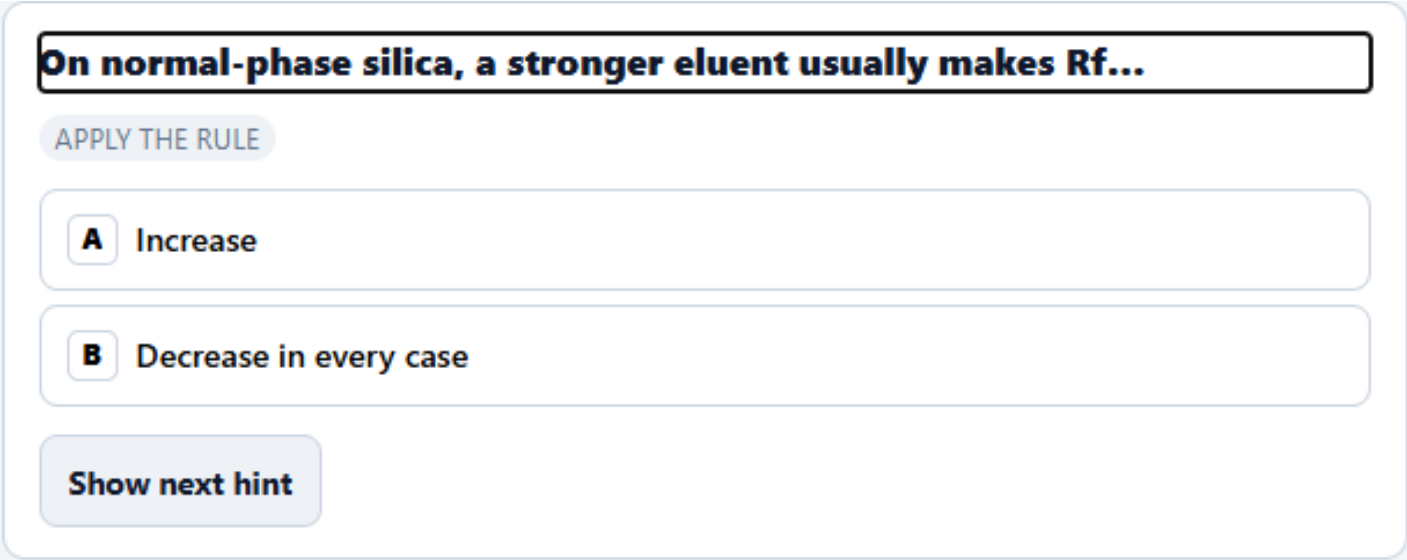
Option 0 (A): Increase

Option 1 (B): Decrease in every case

Screenshot: rf-calculation-stage-2-molecule



Screenshot: rf-calculation-stage-2-question



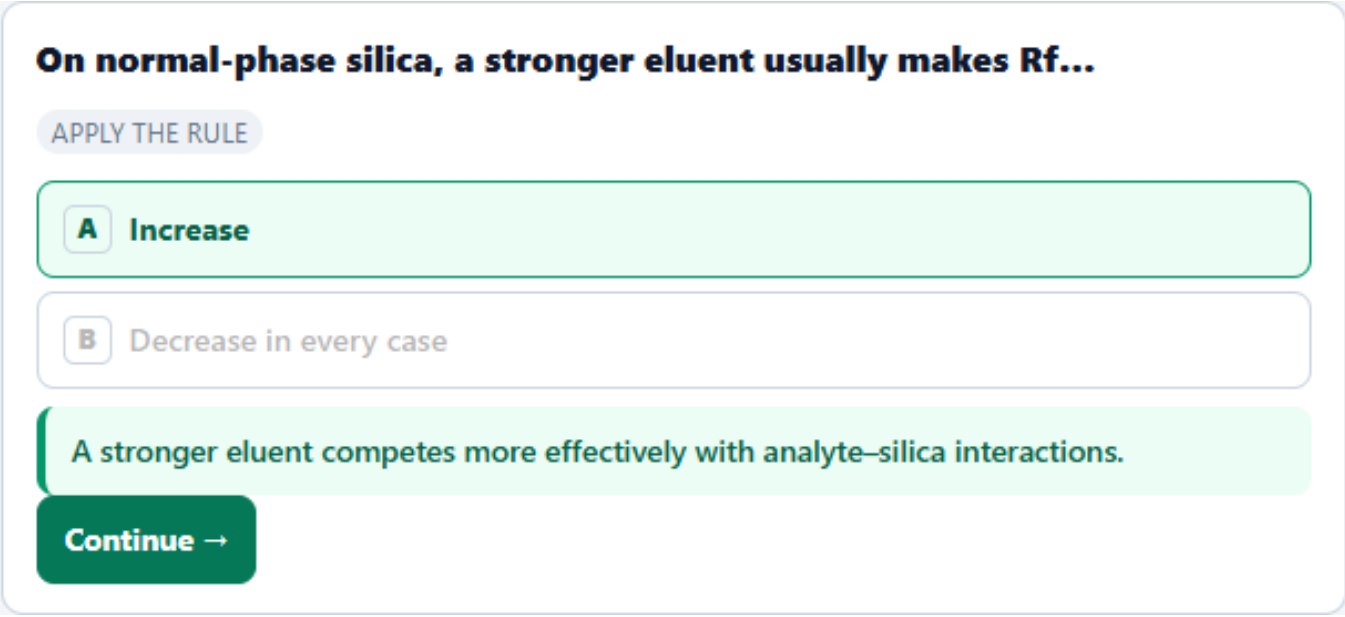
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider the balance between mobile and stationary phase attraction.

Hint 2: More elution usually moves the spot farther.

Explanation: A stronger eluent competes more effectively with analyte–silica interactions.

Screenshot: rf-calculation-stage-2-feedback



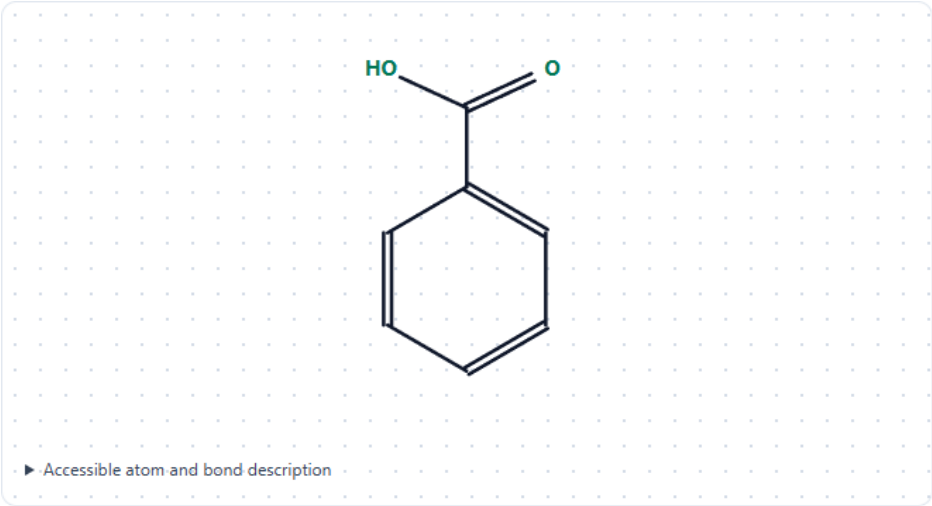
rf-calculation:stage-3 | Two compounds with the same Rf on one plate are...

Answer type: choice

Option 0 (A): Not proven identical

Option 1 (B): Necessarily identical

Screenshot: rf-calculation-stage-3-molecule



Screenshot: rf-calculation-stage-3-question



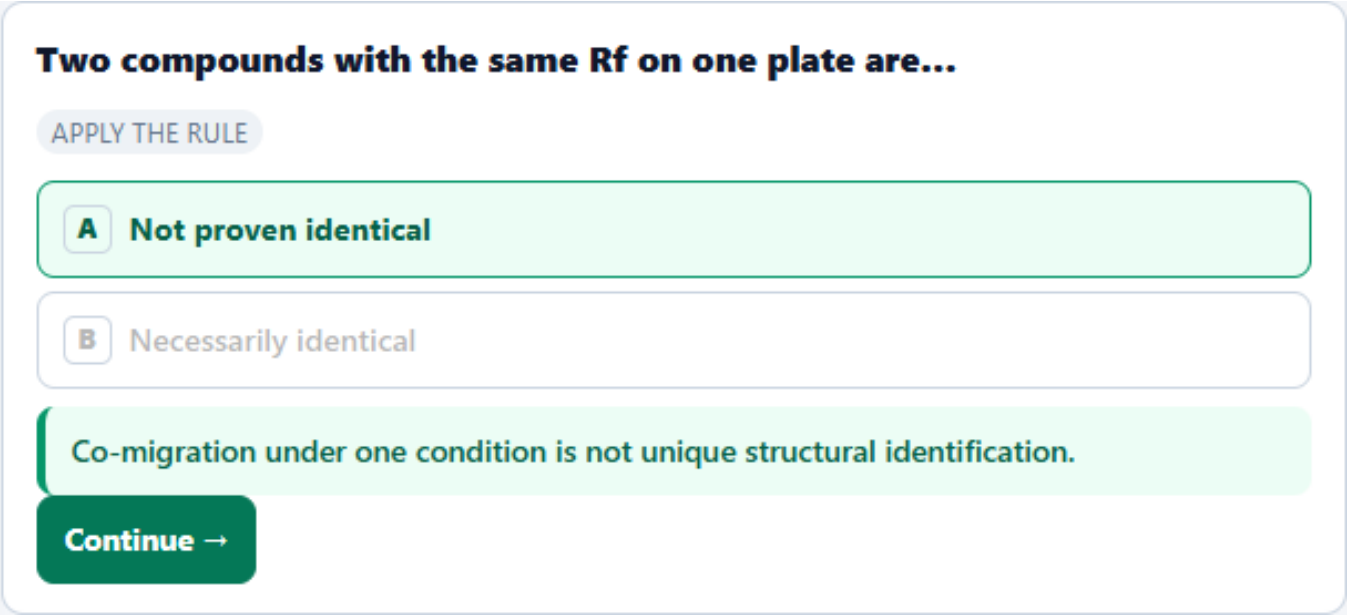
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Different molecules can have similar retention.

Hint 2: Use another condition or independent analytical evidence.

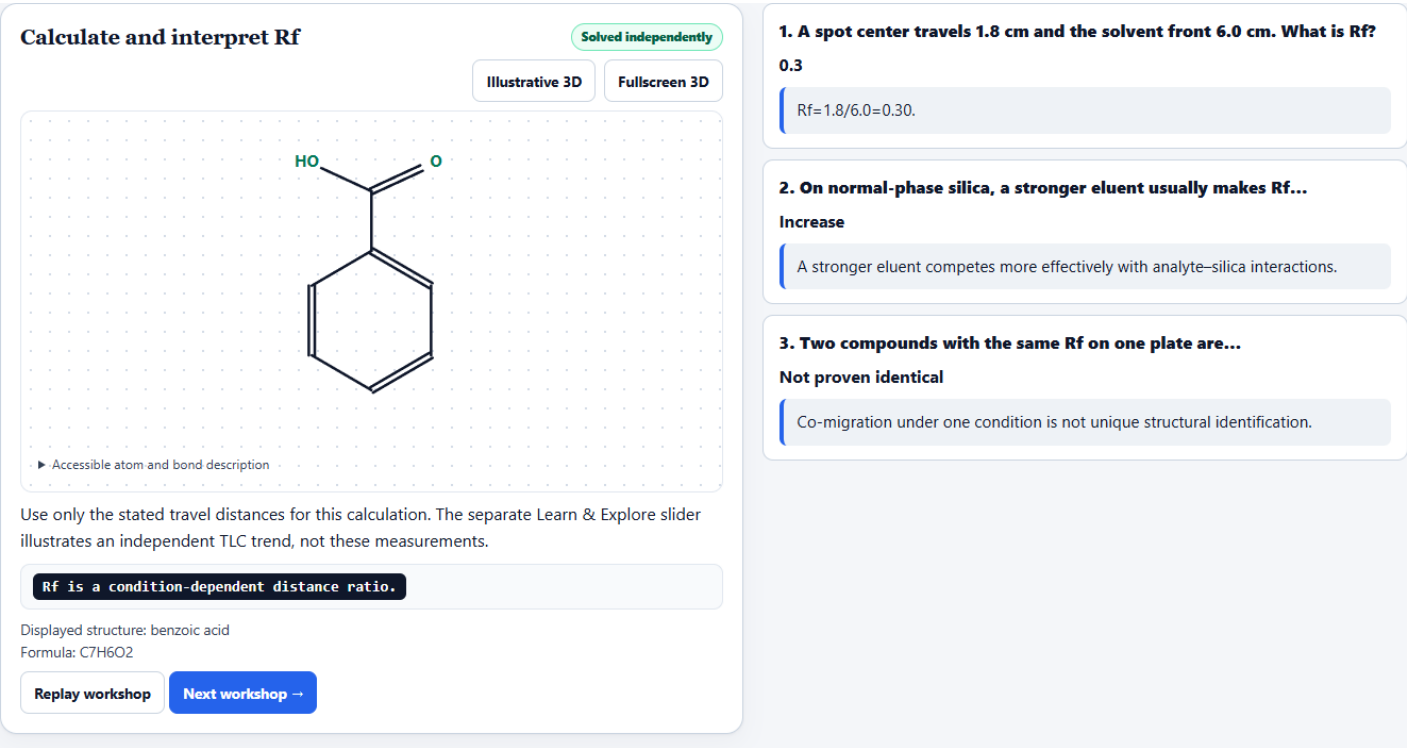
Explanation: Co-migration under one condition is not unique structural identification.

Screenshot: rf-calculation-stage-3-feedback



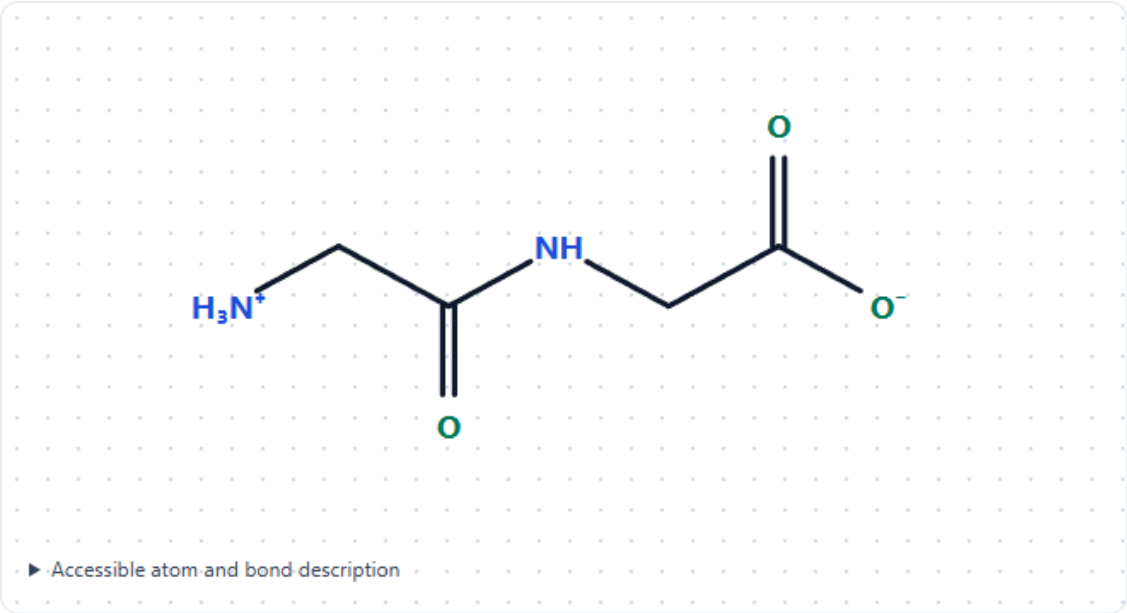
Completion summary: Rf is a condition-dependent distance ratio.

Screenshot: rf-calculation-complete



chromatography-match | Match a protein to a column

Screenshot: chromatography-match-stage-1-molecule



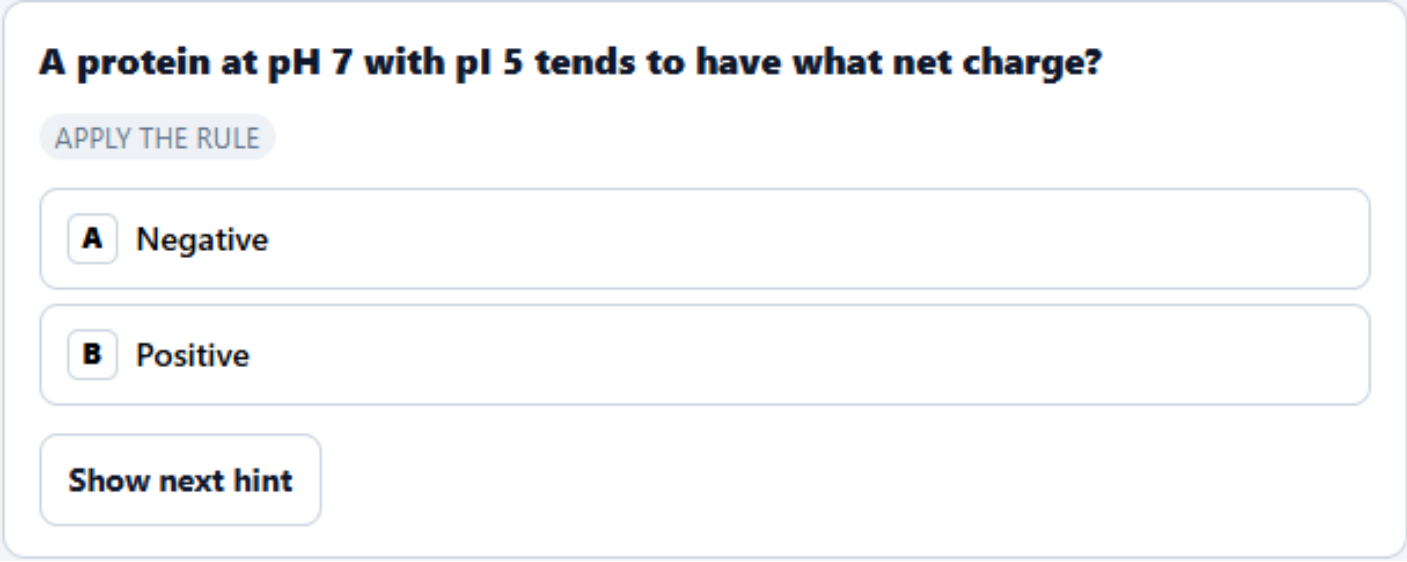
chromatography-match:stage-1 | A protein at pH 7 with pI 5 tends to have what net charge?

Answer type: choice

Option 0 (A): Negative

Option 1 (B): Positive

Screenshot: chromatography-match-stage-1-question



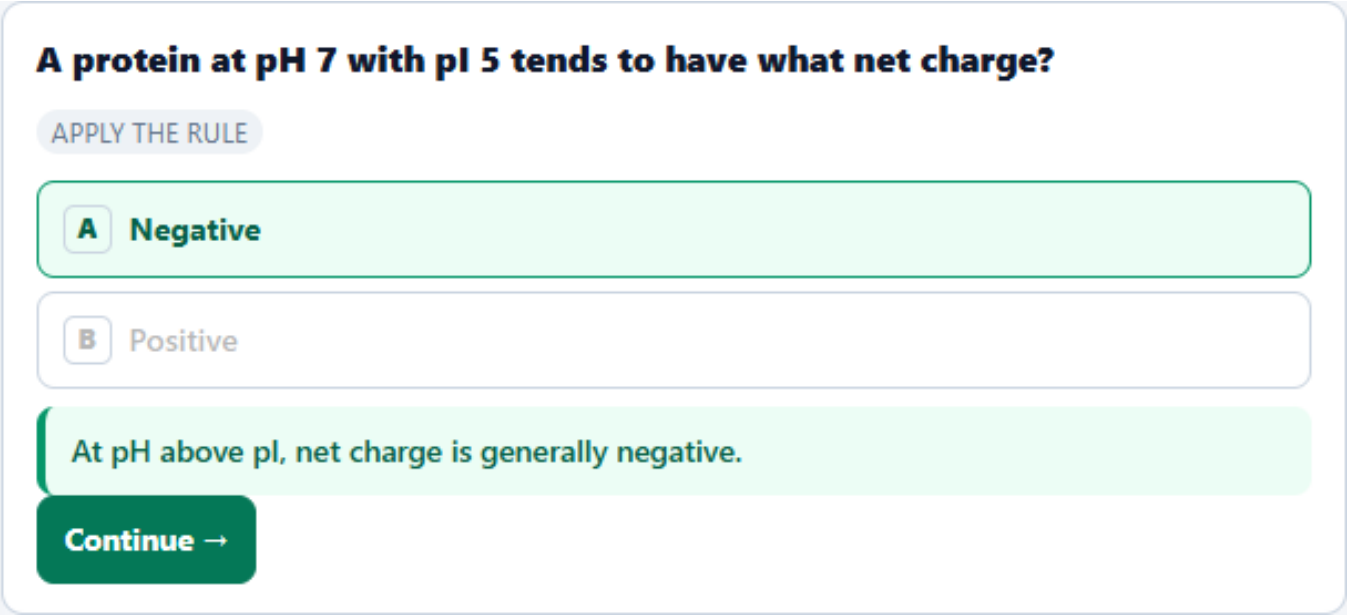
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare pH with pI.

Hint 2: Raising pH removes protons.

Explanation: At pH above pI, net charge is generally negative.

Screenshot: chromatography-match-stage-1-feedback



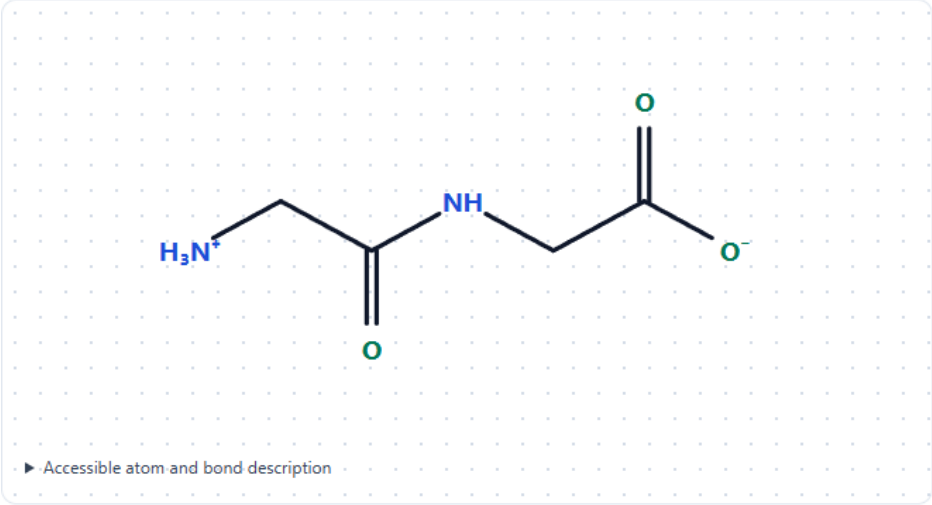
chromatography-match:stage-2 | Which resin would electrostatically bind that protein?

Answer type: choice

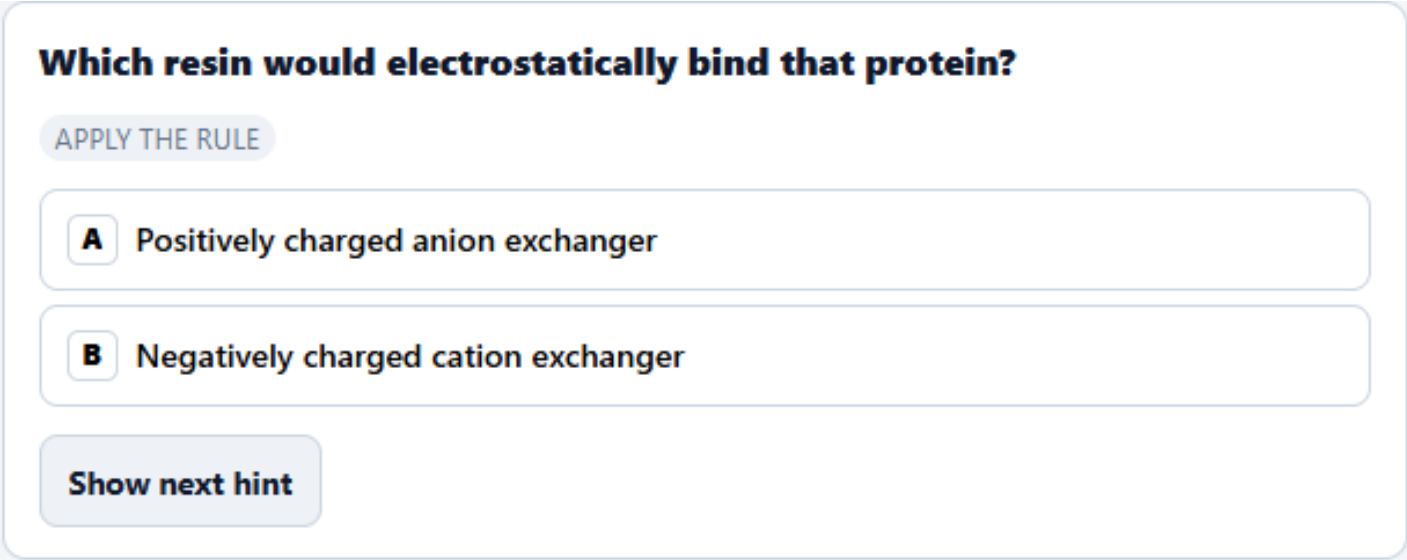
Option 0 (A): Positively charged anion exchanger

Option 1 (B): Negatively charged cation exchanger

Screenshot: chromatography-match-stage-2-molecule



Screenshot: chromatography-match-stage-2-question



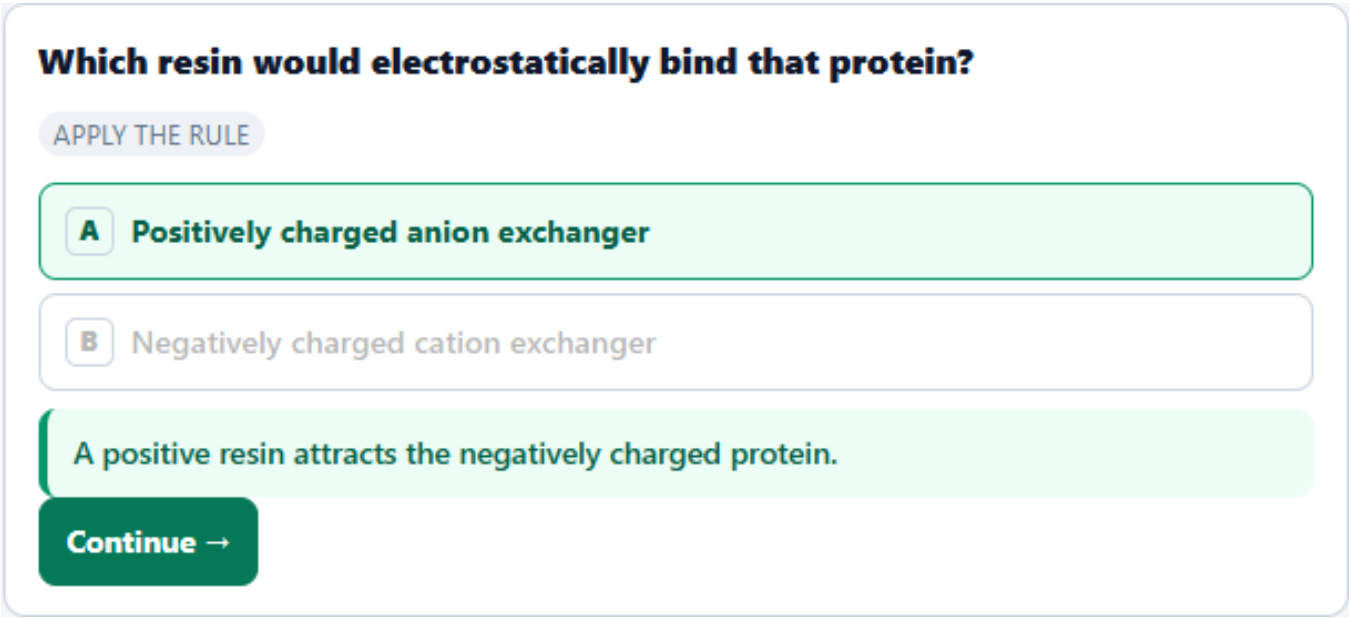
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Opposite charges attract.

Hint 2: Name the exchanger by the ions it binds.

Explanation: A positive resin attracts the negatively charged protein.

Screenshot: chromatography-match-stage-2-feedback



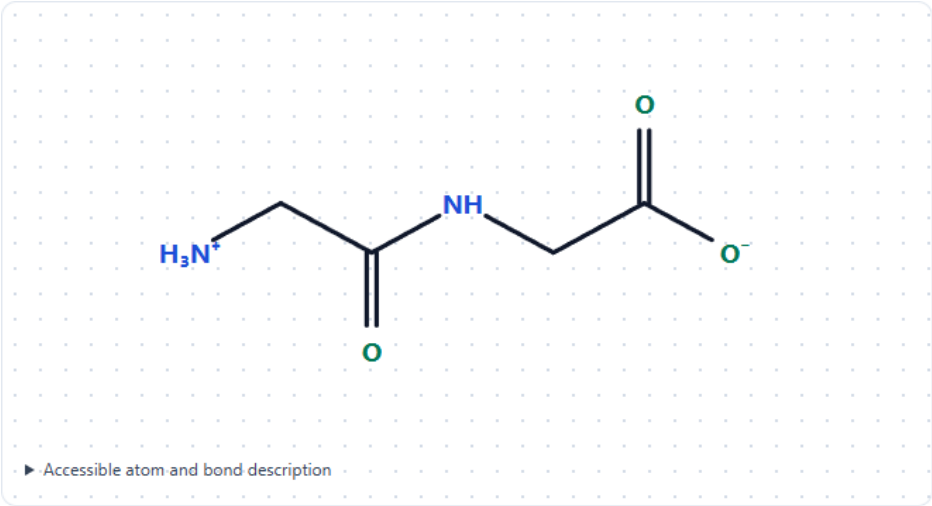
chromatography-match:stage-3 | In ideal size exclusion, larger molecules within the useful separation range usually elute...

Answer type: choice

Option 0 (A): Earlier

Option 1 (B): Later

Screenshot: chromatography-match-stage-3-molecule



Screenshot: chromatography-match-stage-3-question

In ideal size exclusion, larger molecules within the useful separation range usually elute...

APPLY THE RULE

A Earlier

B Later

Show next hint

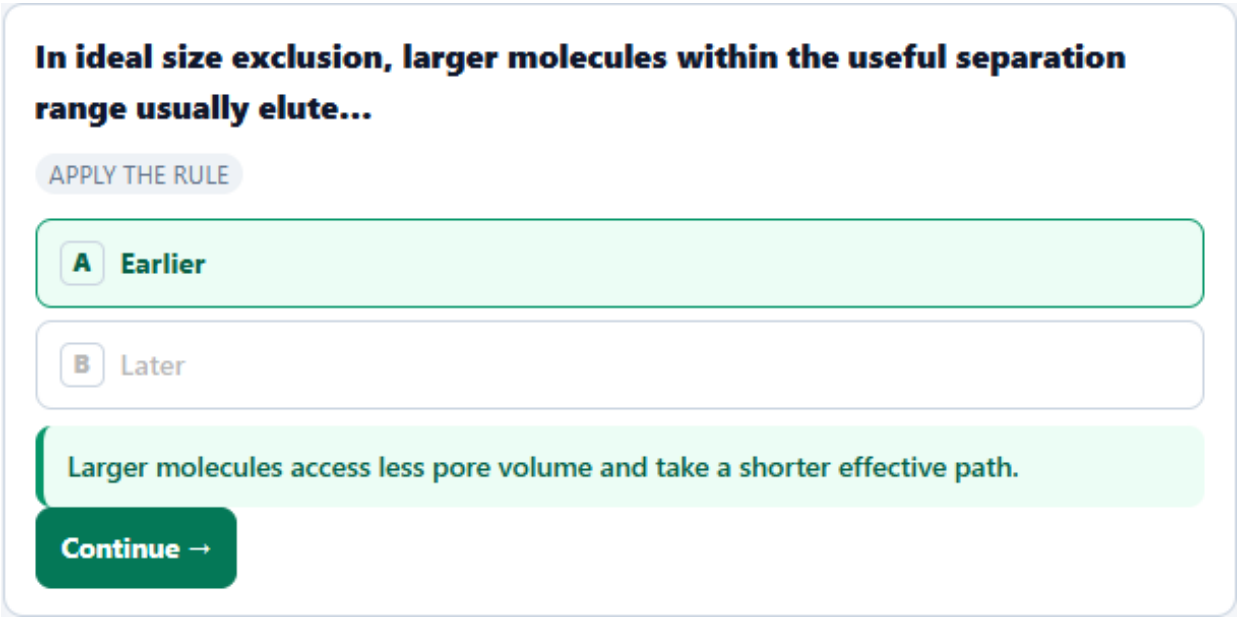
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Smaller molecules explore more pores.

Hint 2: Pore entry delays elution.

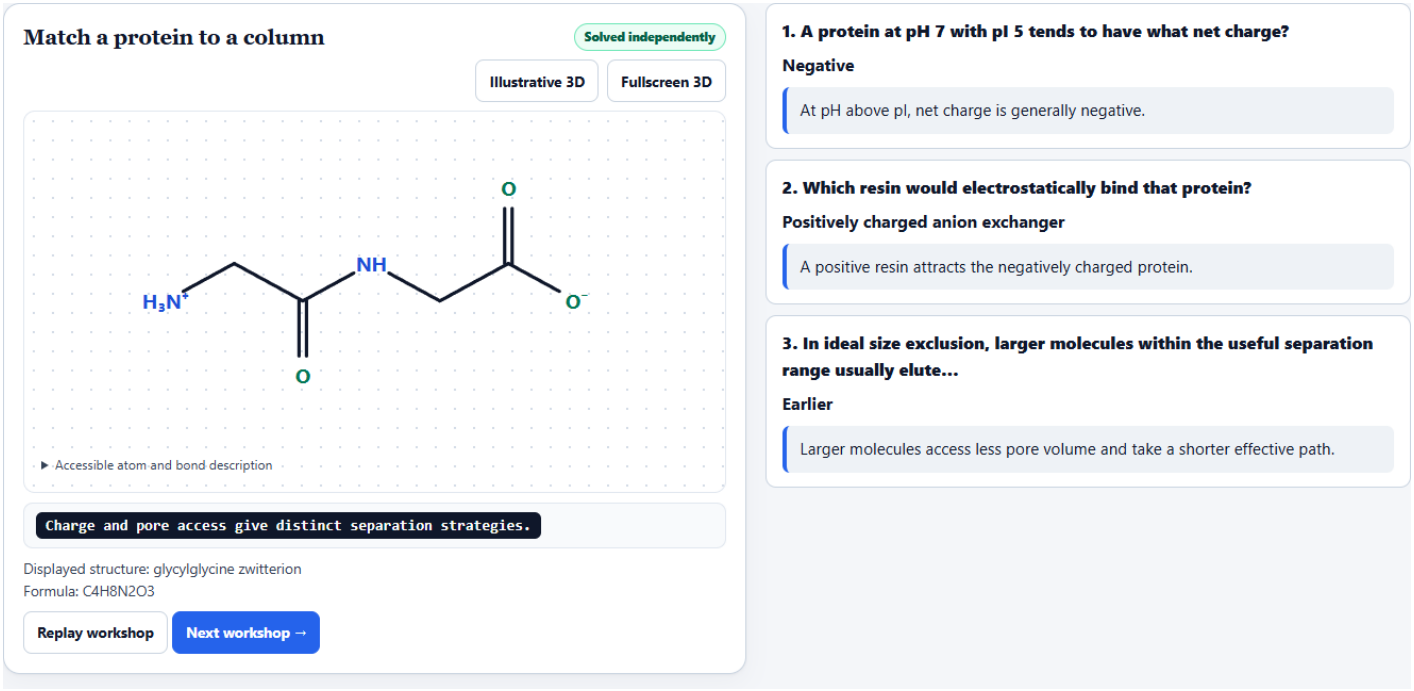
Explanation: Larger molecules access less pore volume and take a shorter effective path.

Screenshot: chromatography-match-stage-3-feedback



Completion summary: Charge and pore access give distinct separation strategies.

Screenshot: chromatography-match-complete



Molecular repository: author claims to verify

benzoicacid

Author claims: benzoic acid | formula C7H6O2 | canonical composition C7H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

benzoate

Author claims: benzoate | formula C7H5O2 | canonical composition C7H5O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

toluene

Author claims: methylbenzene (toluene) | formula C7H8 | canonical composition C7H8 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=93.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=127.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=195.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=229.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=195.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=127.00; charge=0; explicit H override=none

6 (G): C; x=170.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1]]

glycylglycine

Author claims: glycylglycine zwitterion | formula C4H8N2O3 | canonical composition C4H8N2O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

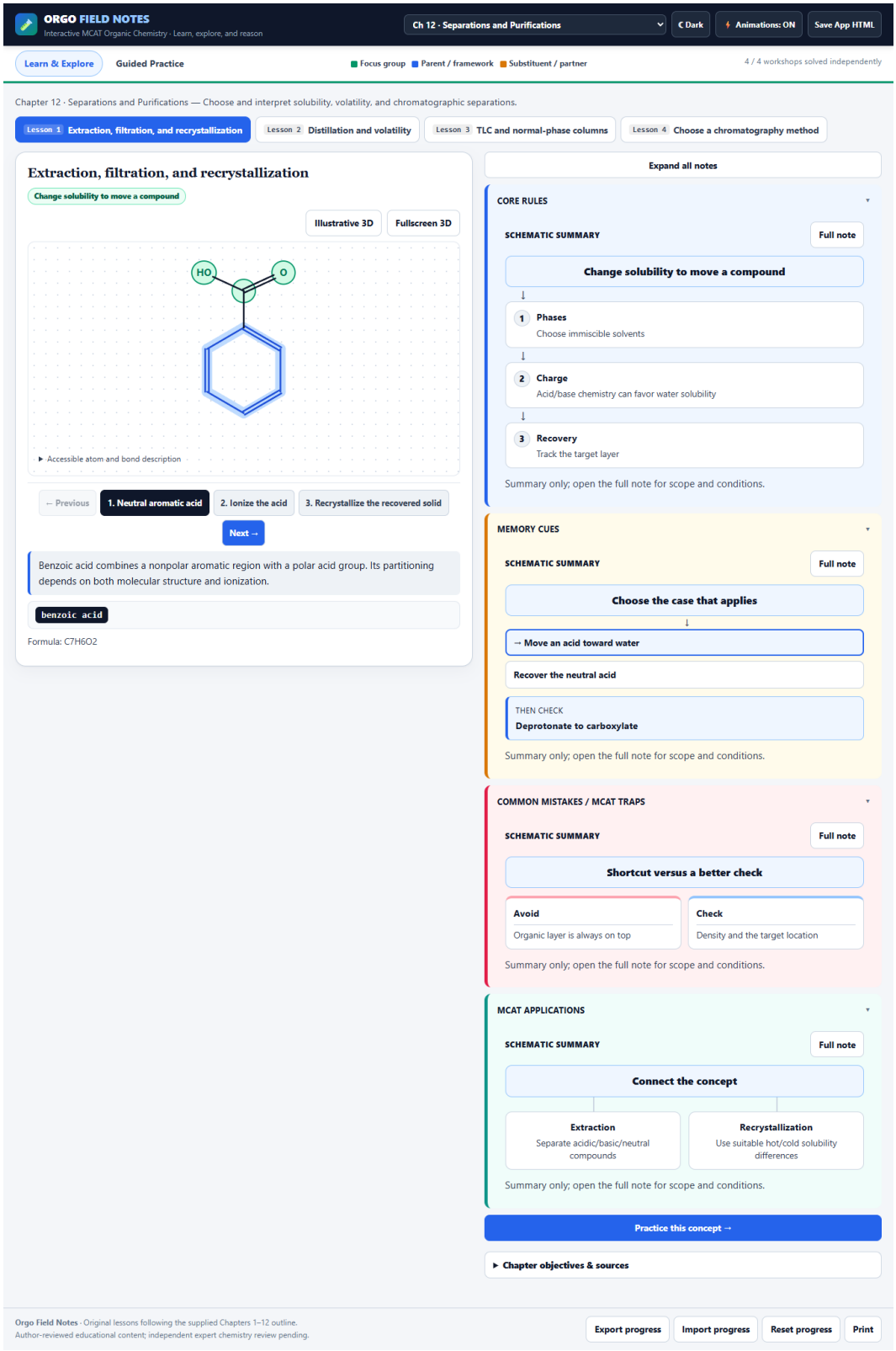
8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

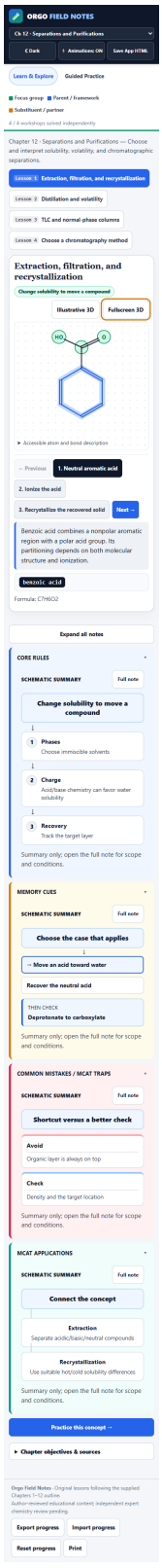
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

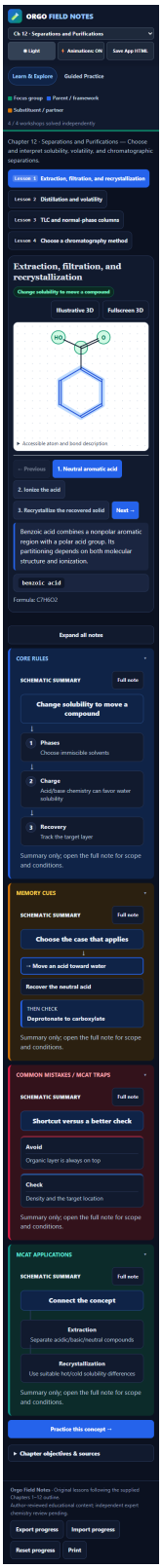
Screenshot: layout-desktop-light



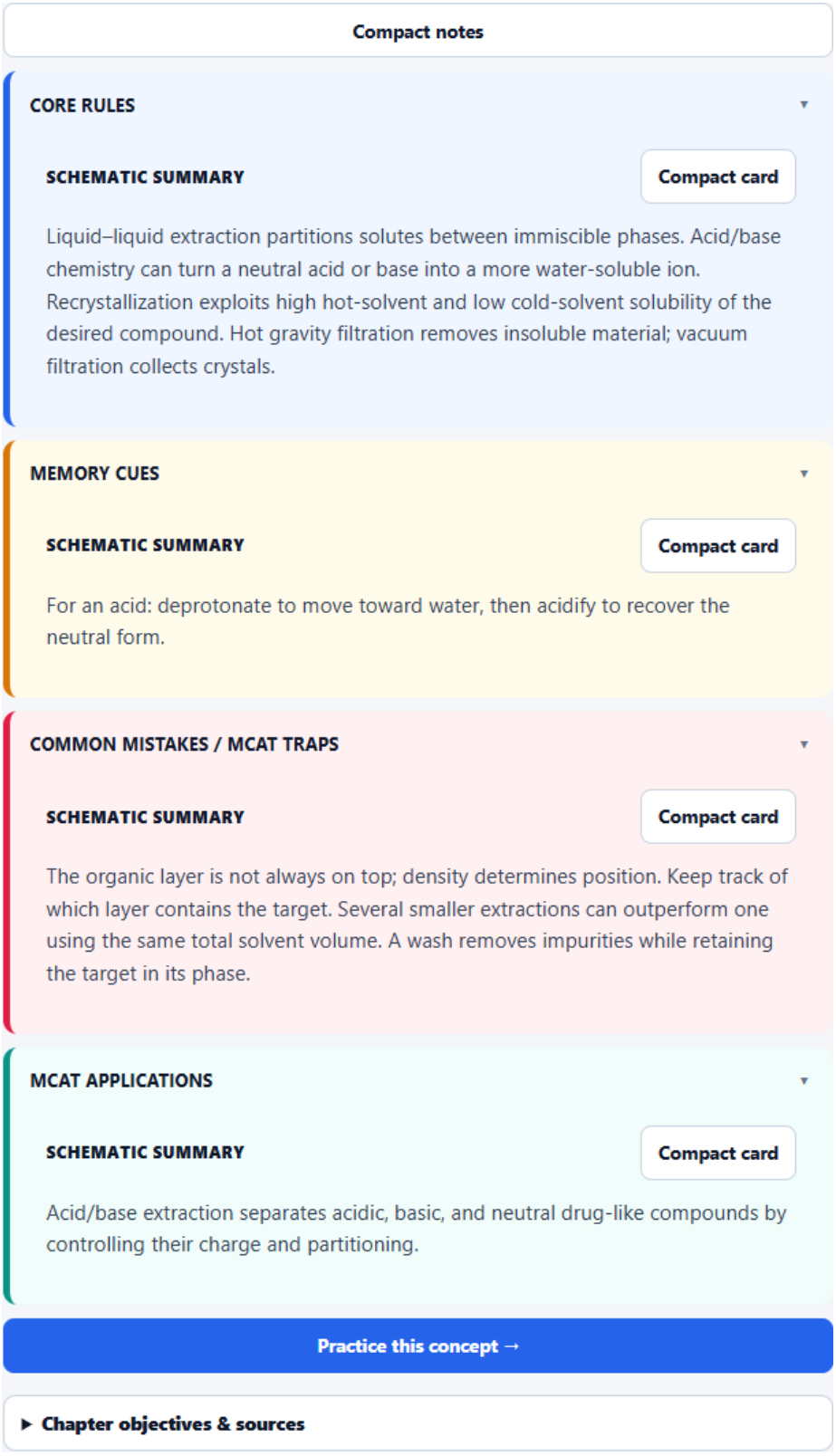
Screenshot: layout-mobile-light



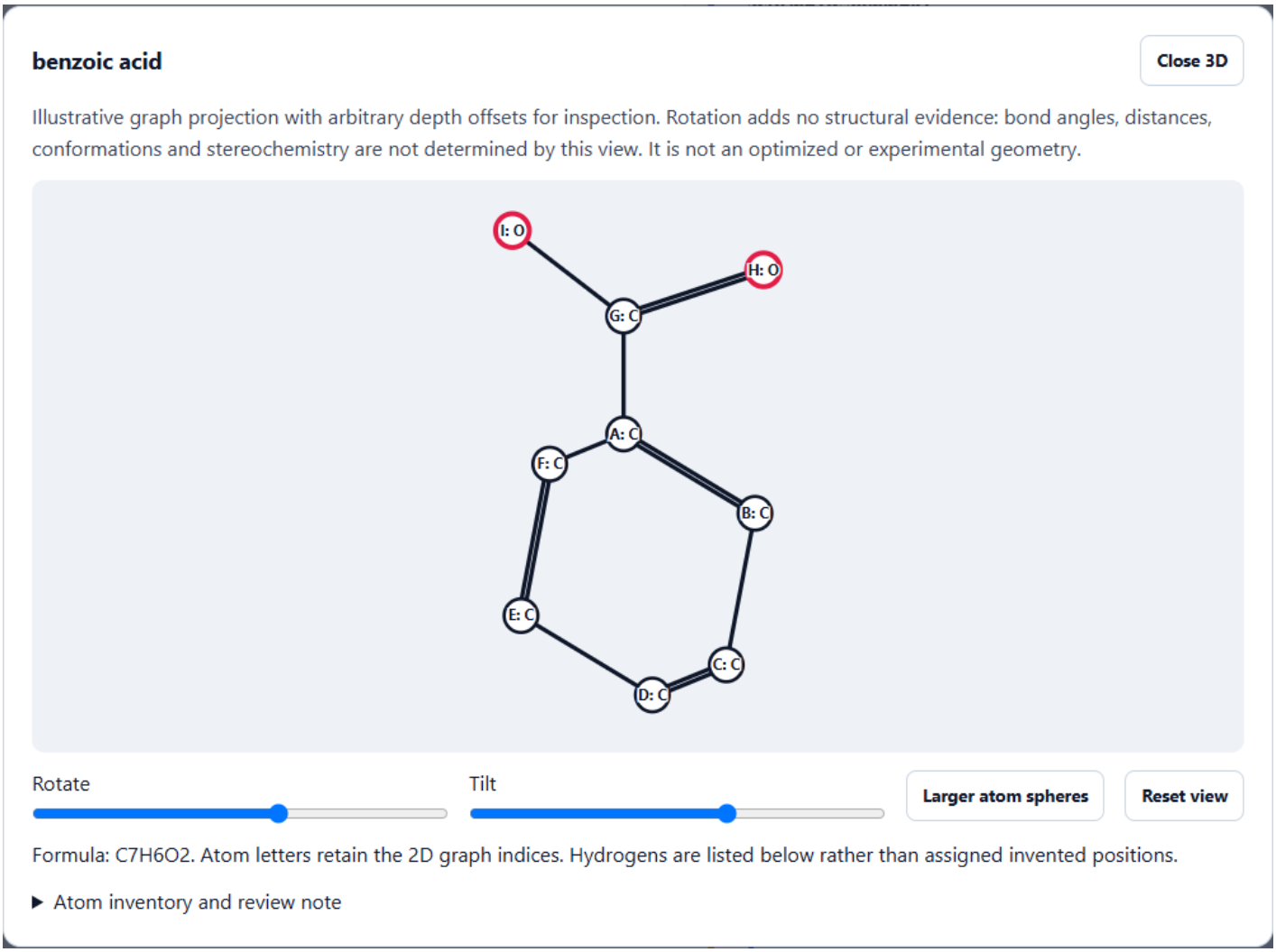
Screenshot: layout-mobile-dark



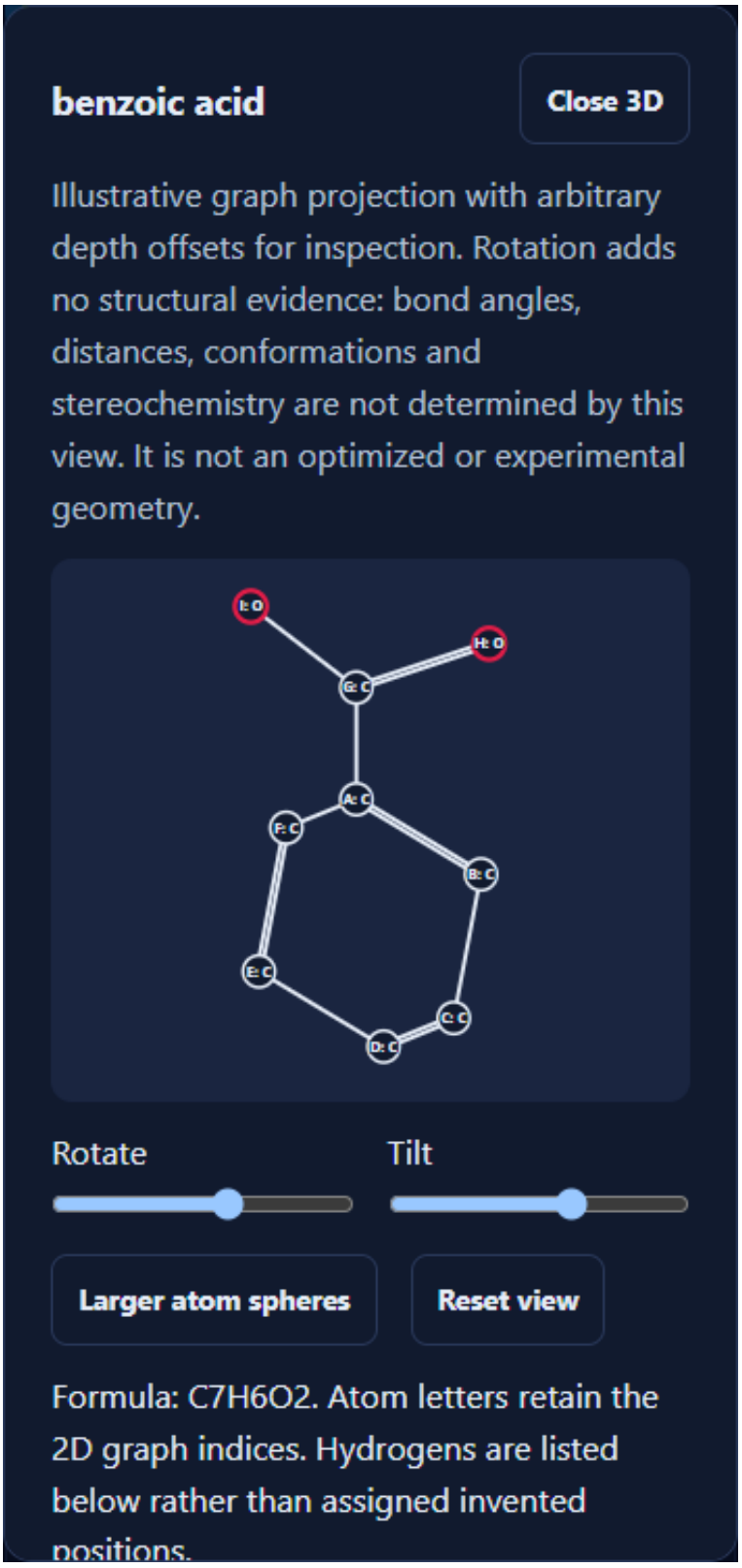
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

60 screenshots and 12 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.